CONSTRUCTION 1

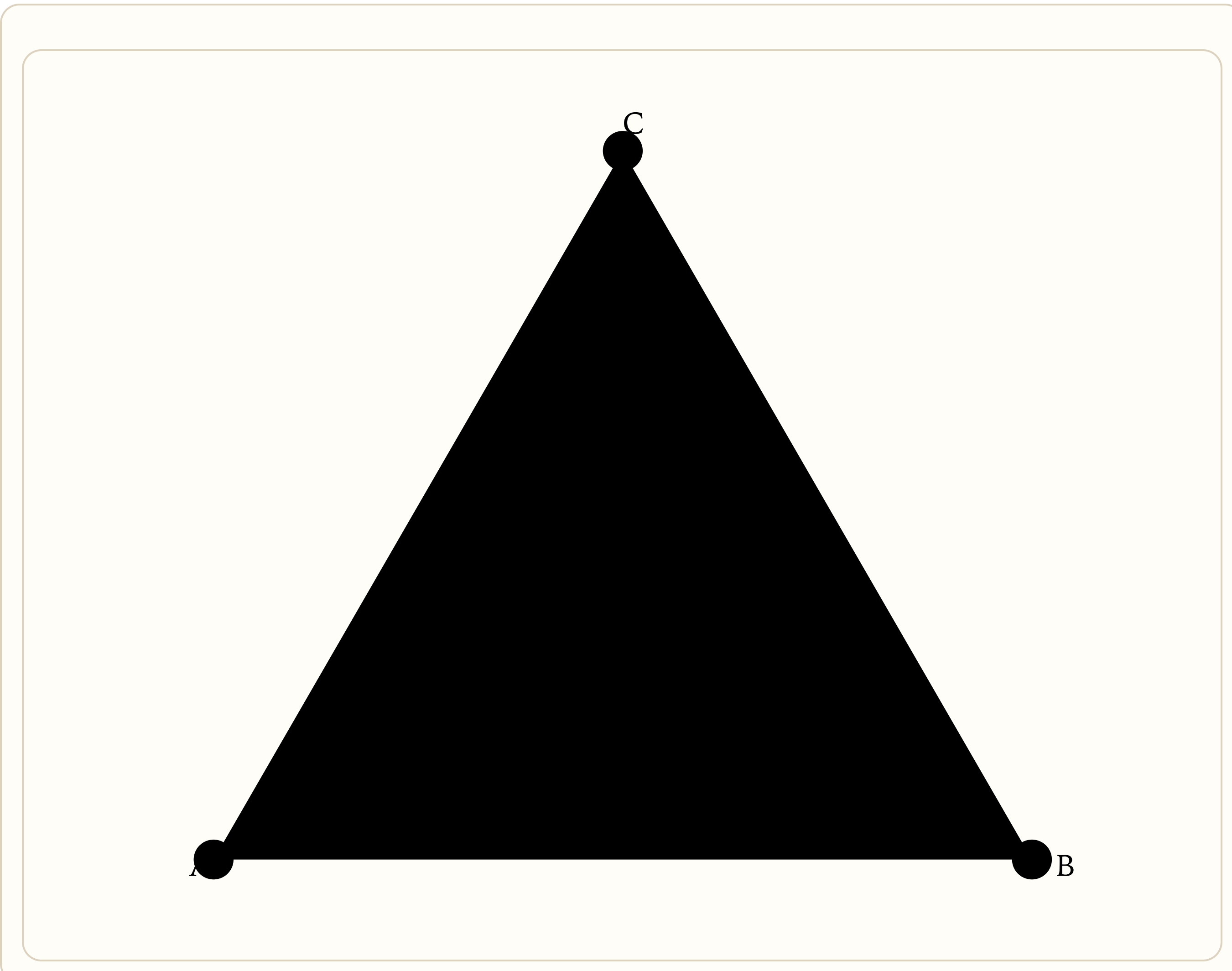
How to make an equilateral triangle

STATEMENT

Given a straight line AB, construct an equilateral triangle on it so that AB is the base.

GIVEN

Draw circle X around point A with radius AB.



CONSTRUCTION 1A

How to place at a given point a line equal to a given line

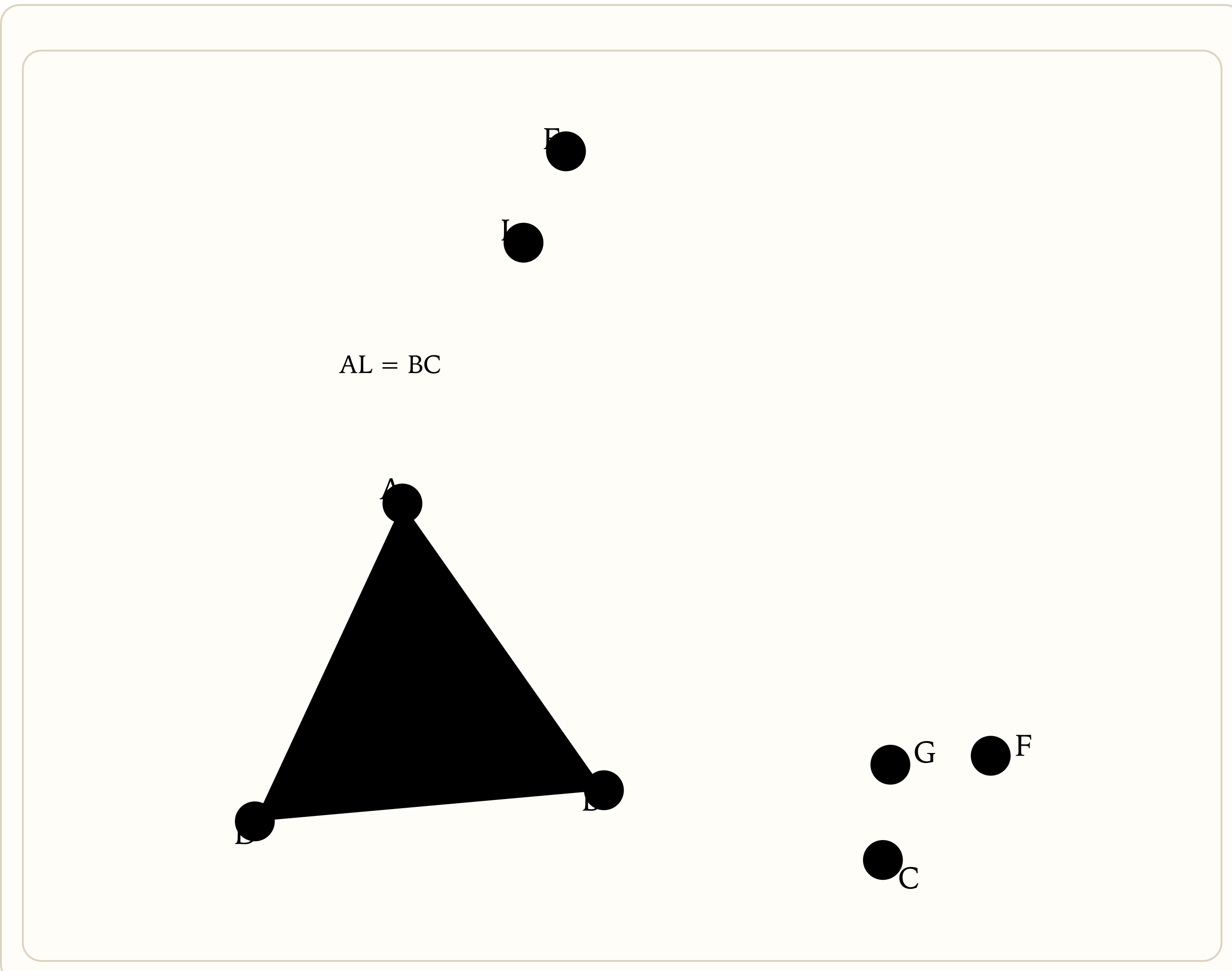
Euclid, Elements I.2

STATEMENT

If someone gives us a point A and a straight line BC, how can we draw from the point A a straight line equal to BC?

GIVEN

Join the points A and B with a straight line.



CONSTRUCTION 1B

How to cut off from the greater of two lines a line equal to the less

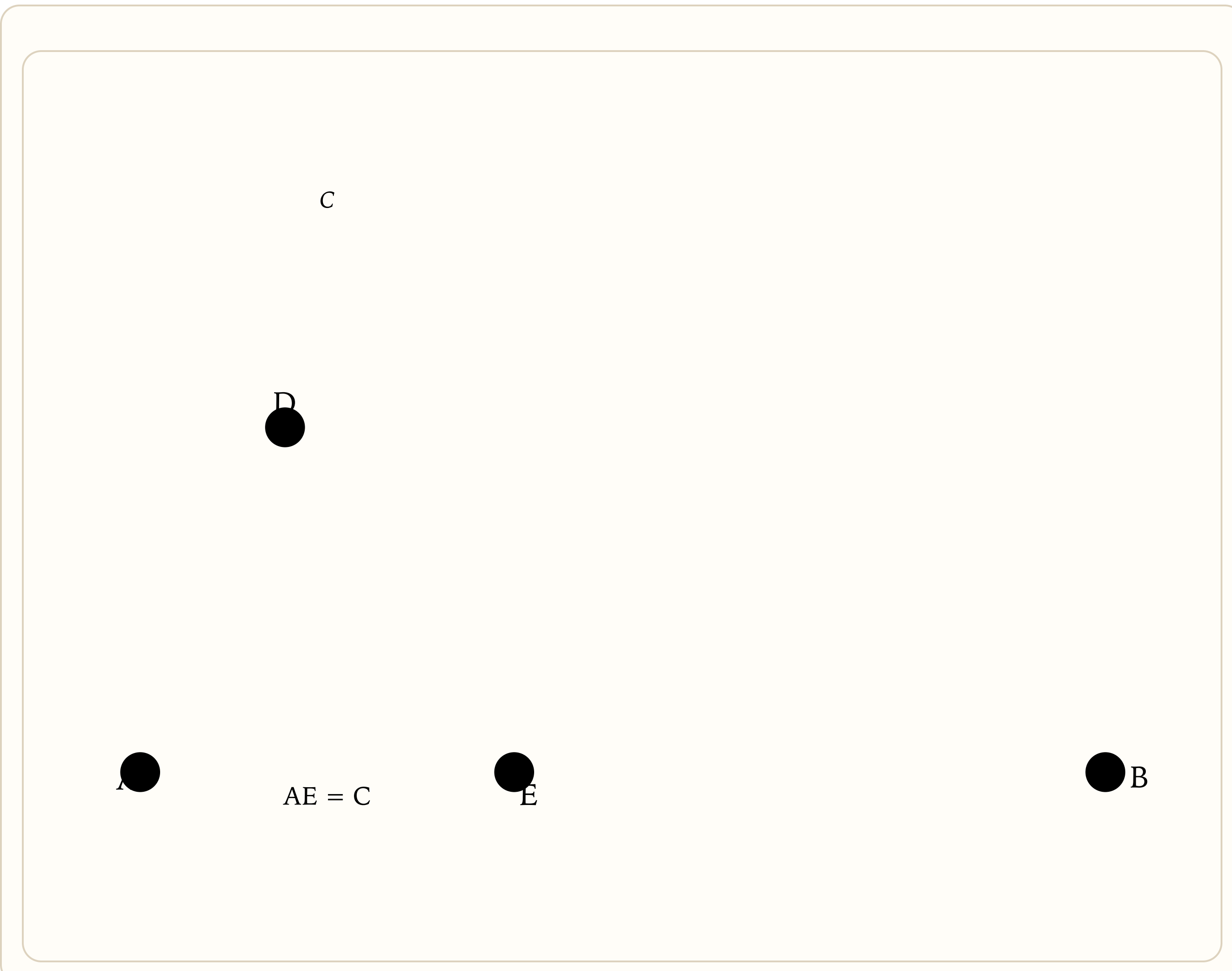
Euclid, Elements I.3

STATEMENT

Given two unequal straight lines AB and C, cut off from the greater one AB a straight line equal to the less one C.

GIVEN

At the point A place a straight line AD equal to C.



THEOREM 2

Side-Angle-Side (SAS)

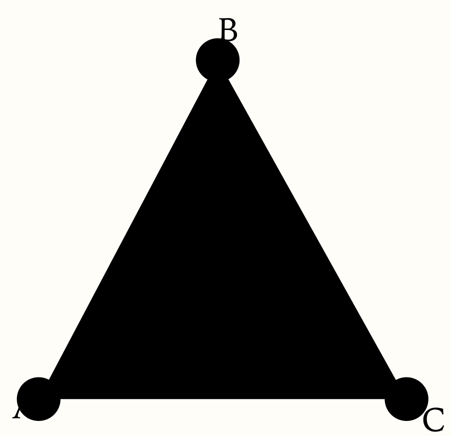
SAS Congruence

STATEMENT

If in one triangle a side, the next angle, and the next side are respectively equal to a side, the next angle, and the next side in another triangle, then all the corresponding sides and angles of the two triangles are equal, and they have equal areas.

GIVEN

Imagine $\triangle ABC$ and $\triangle DEF$ with $AB = DE$, $\angle ABC = \angle DEF$, and $BC = EF$.



THEOREM 3

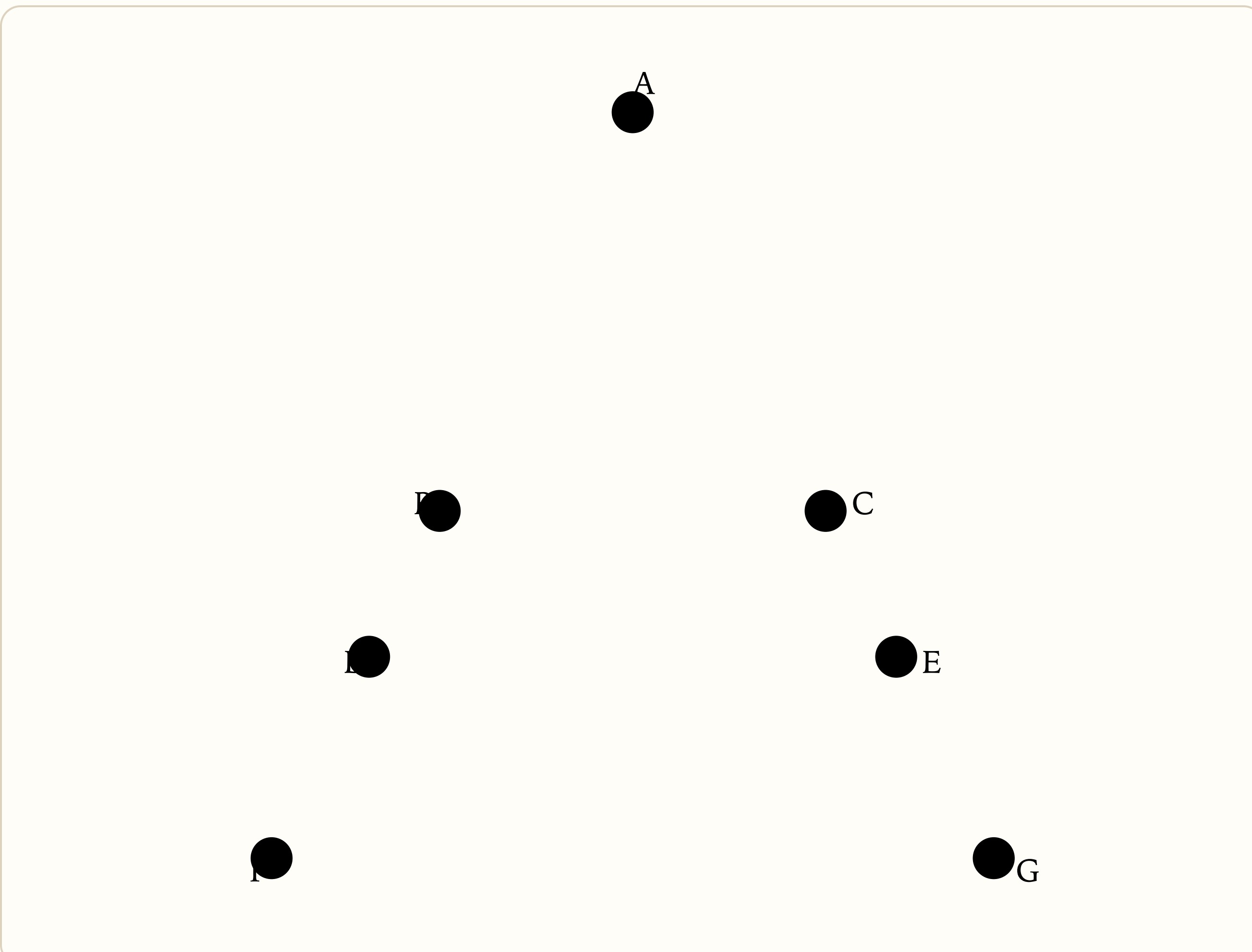
Base angles of an isosceles triangle

STATEMENT

In an isosceles triangle, the base angles are equal to each other, and the angles under the base are equal to each other.

GIVEN

Let $\triangle ABC$ be isosceles with $AB = AC$. We claim $\angle ABC = \angle ACB$, and (extending the equal sides past the base to F and G) the angles under the base are equal too. Pick D at random on BF.



THEOREM 4

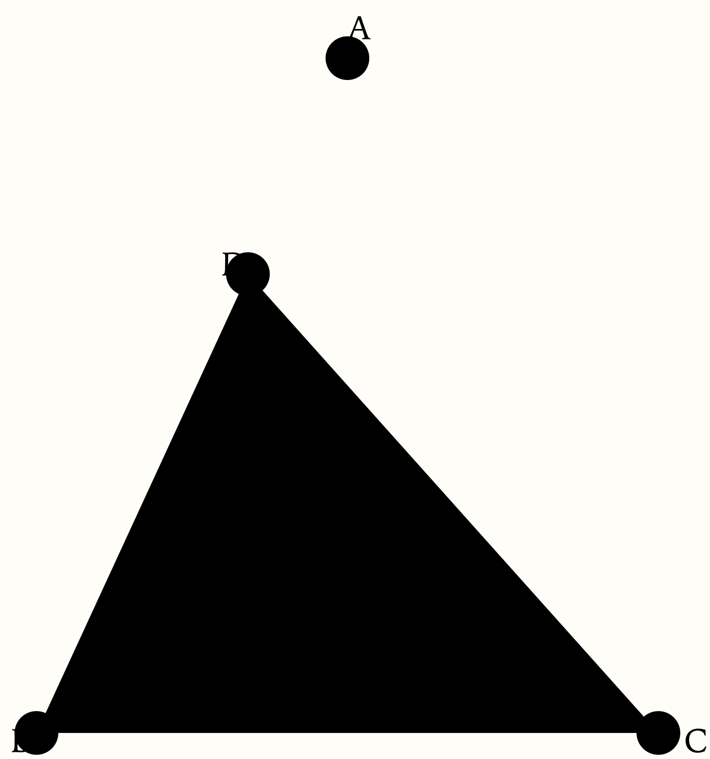
Converse of the isosceles base-angle theorem

STATEMENT

If two angles in a triangle are equal to each other, then the sides opposite them are also equal — the triangle is isosceles (at least two sides equal).

GIVEN

In △ABC suppose ∠ABC = ∠ACB. First take side AB, the base BC, and the angle they contain, ∠ABC.



△DBC is a part of △ACB, yet equal to it — absurd

THEOREM 5

Two triangles cannot stand on the same base with the same sides

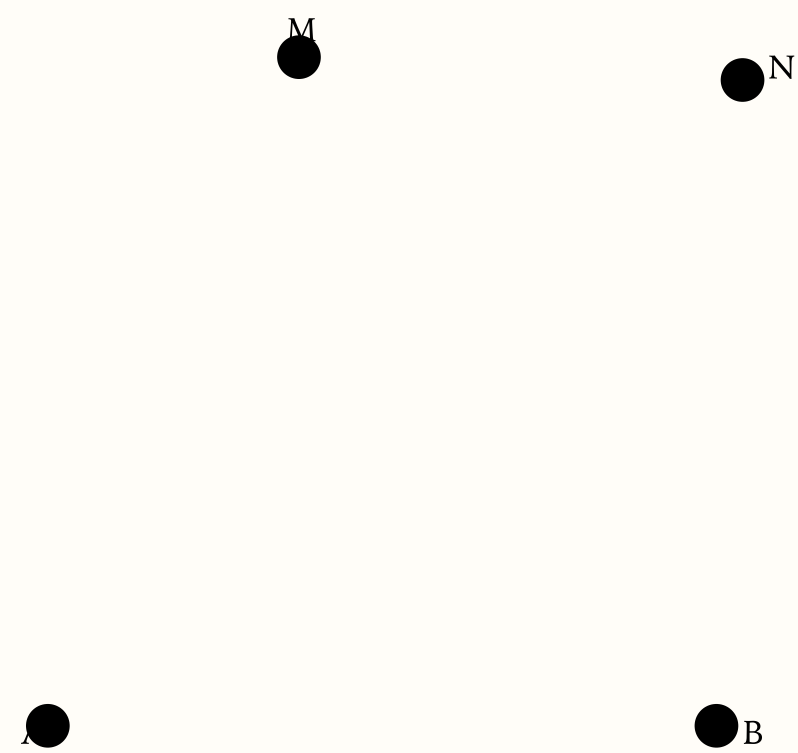
Triangles are rigid

STATEMENT

If two triangles stand on the same base and on the same side of it, they cannot have the two sides drawn from one end of the base equal to each other, and the two sides drawn from the other end equal to each other, and yet be different triangles.

GIVEN

A square of four hinged rods can be kicked over into a slanted shape, keeping every rod the same length. Can a triangle do the same?



(the figure cannot really exist — that is what we prove)

Side-Side-Side (SSS)

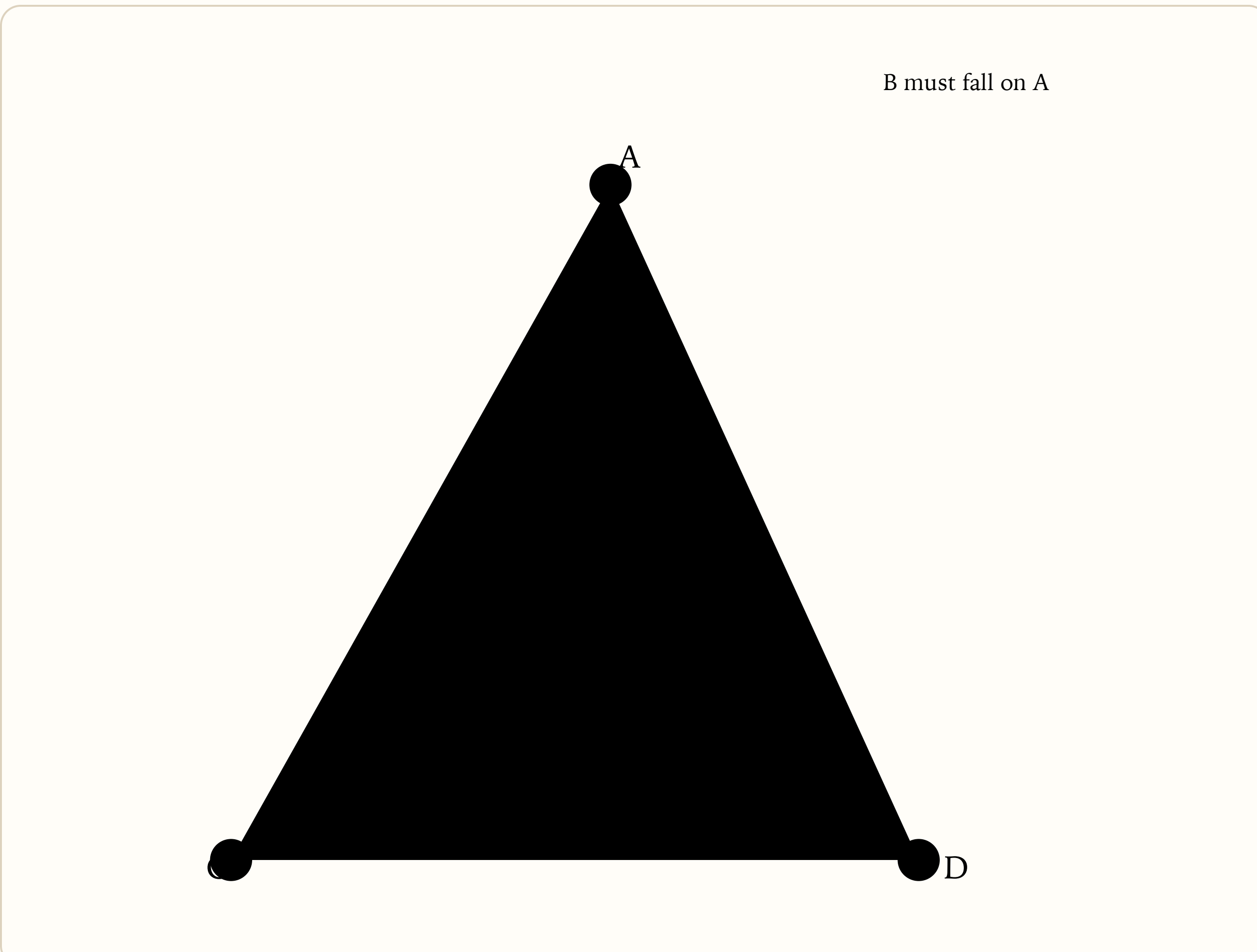
SSS Congruence

STATEMENT

If the three sides of one triangle are equal to the three corresponding sides of another triangle, then the two triangles also have their corresponding angles equal (those between equal sides), and equal areas.

GIVEN

Let △A and △B have three pairs of equal corresponding sides. Place them on the same base CD with equal sides sharing endpoints C and D, so CA = CB and DA = DB.



CONSTRUCTION 7

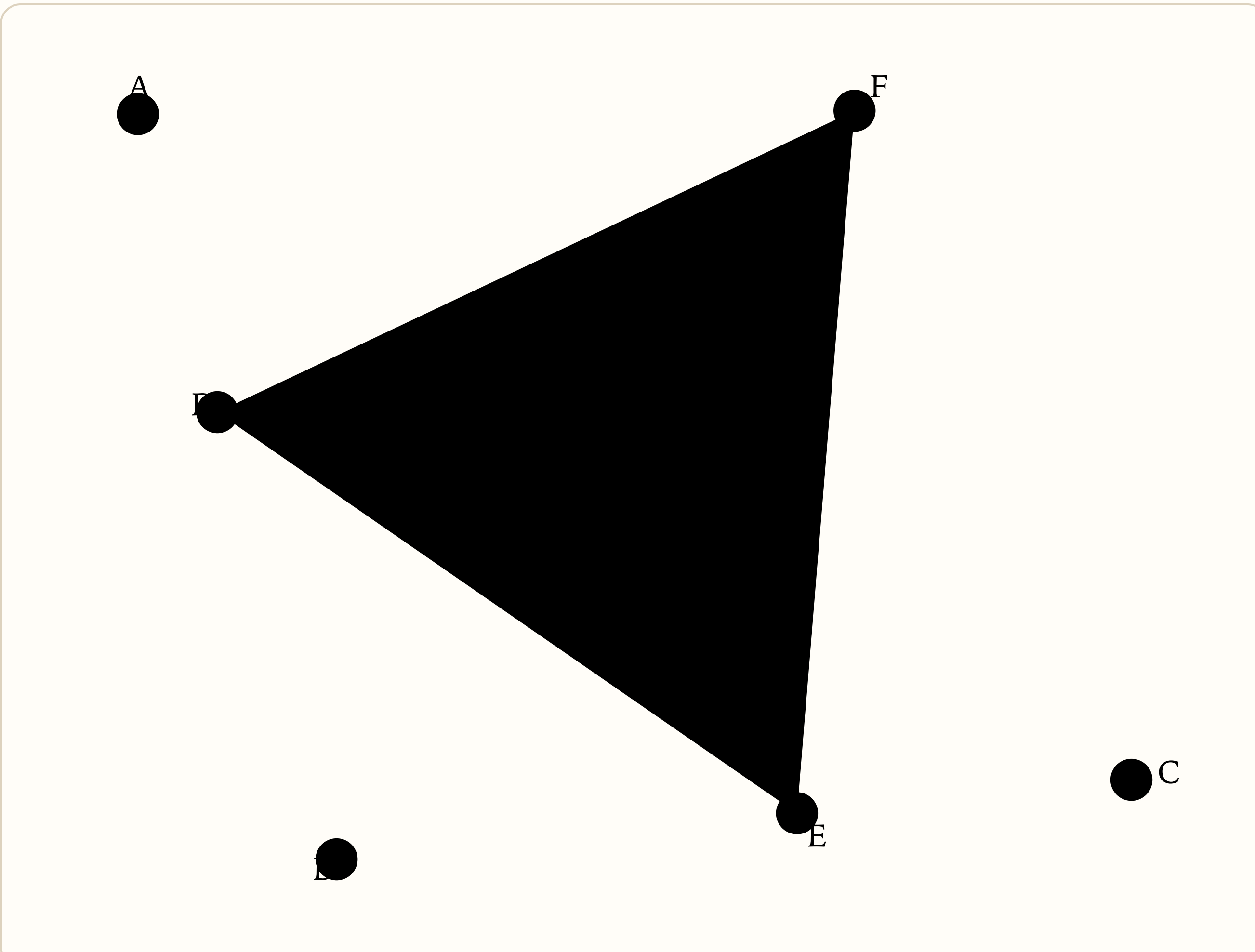
How to bisect any rectilineal angle

STATEMENT

Given any rectilineal angle ABC, construct the ray that bisects it.

GIVEN

Pick any point D on AB, and draw a circle around B with radius BD, cutting off BE = BD.



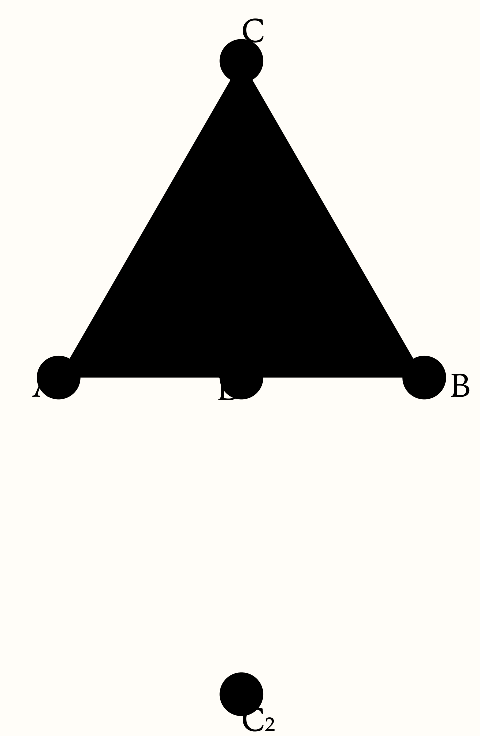
How to bisect any straight line

STATEMENT

Given a straight line AB, construct its midpoint.

GIVEN

Make equilateral triangle ABC on AB.



the two circles cross above and below; the line joining the crossings cuts AB in half

CONSTRUCTION 9

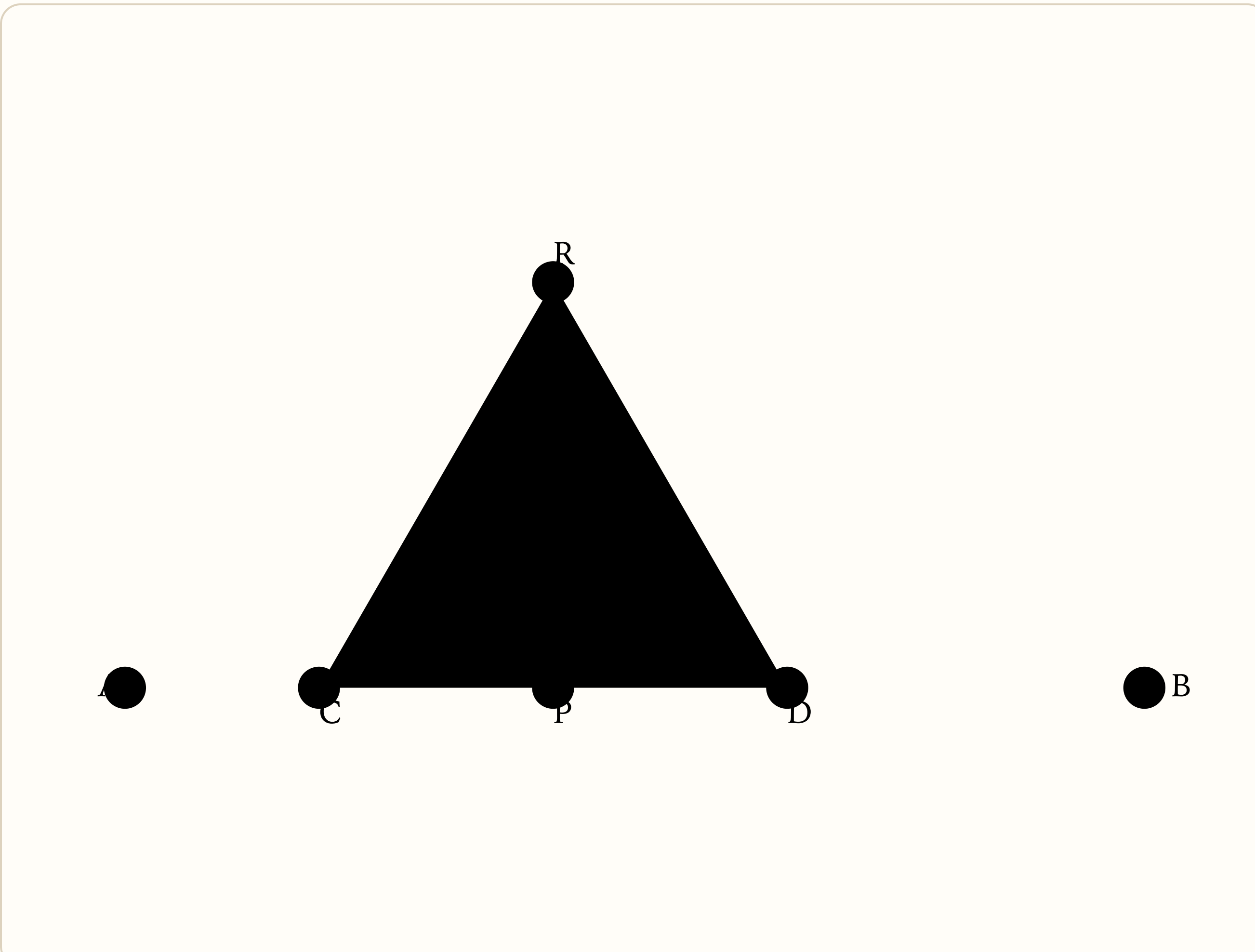
Erect a perpendicular from a point on a line

STATEMENT

Given a straight line AB and a point P on it, draw a line from P at right angles to AB.

GIVEN

Pick any point C on AP; draw a circle around P with radius PC, cutting off PD = PC.



CONSTRUCTION 10

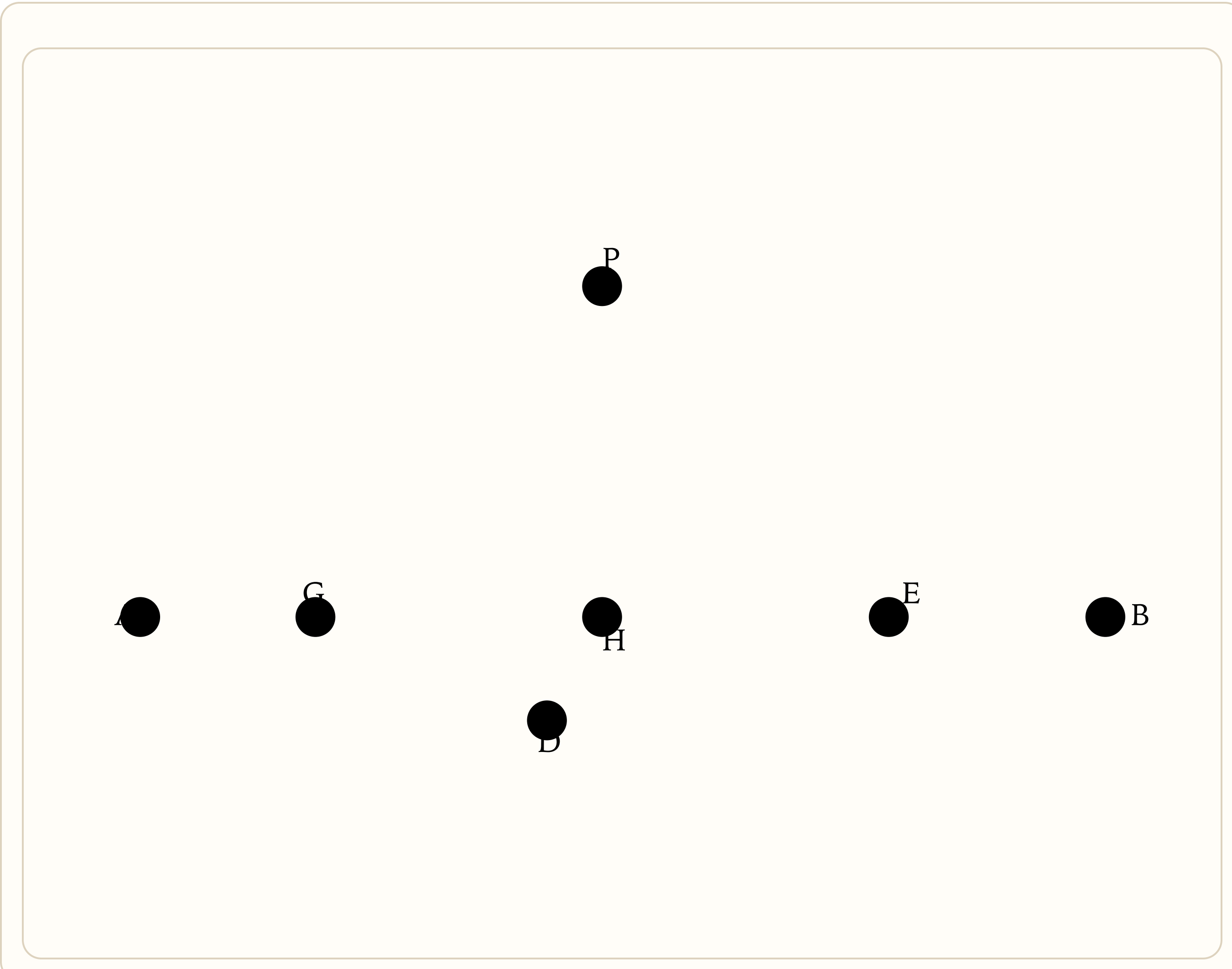
Drop a perpendicular from a point to a line

STATEMENT

Given a straight line AB and a point P not on it, drop a perpendicular from P to AB.

GIVEN

Choose any point D on the opposite side of AB from P.



THEOREM 11

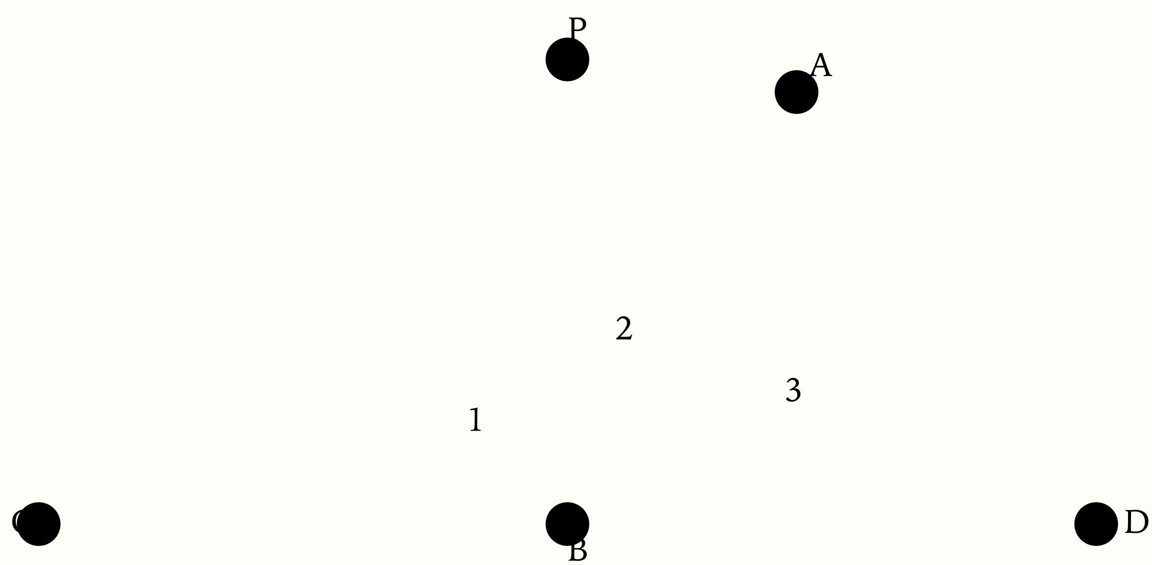
Adjacent angles on a straight line

STATEMENT

When one straight line stands on another, the adjacent angles add up to two right angles.

GIVEN

If PB is perpendicular to CD, clearly ∠PBC + ∠PBD = two right angles. Now let AB stand on CD not necessarily perpendicularly.



∠ABC + ∠ABD = two right angles

THEOREM 12

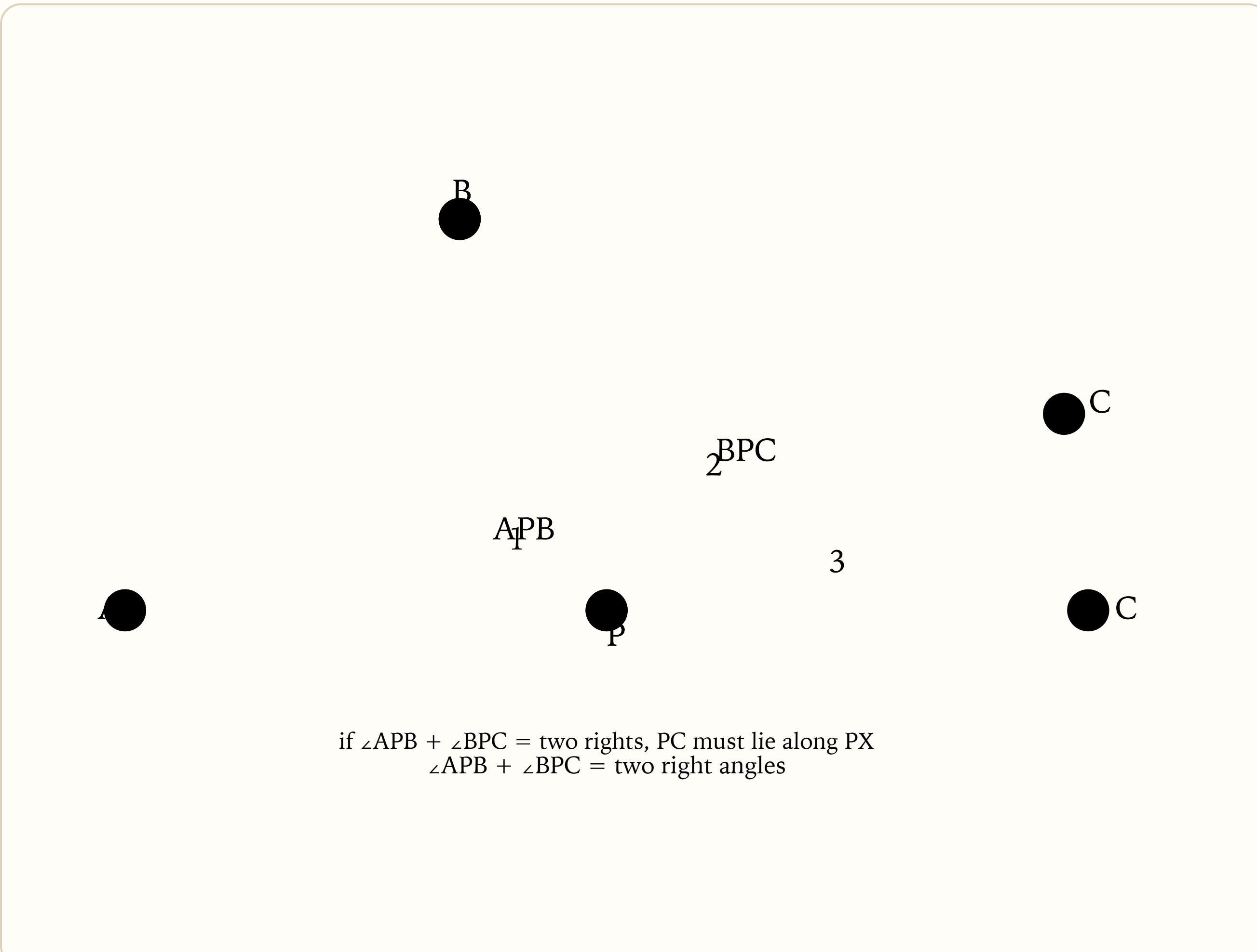
Converse: angles sum to two rights ⇒ straight line

STATEMENT

If two adjacent angles add up to two right angles, then the outer legs form one straight line.

GIVEN

Let ∠APB + ∠BPC = two right angles. Claim: A, P, C are collinear (AP and PC form a straight line).



THEOREM 13

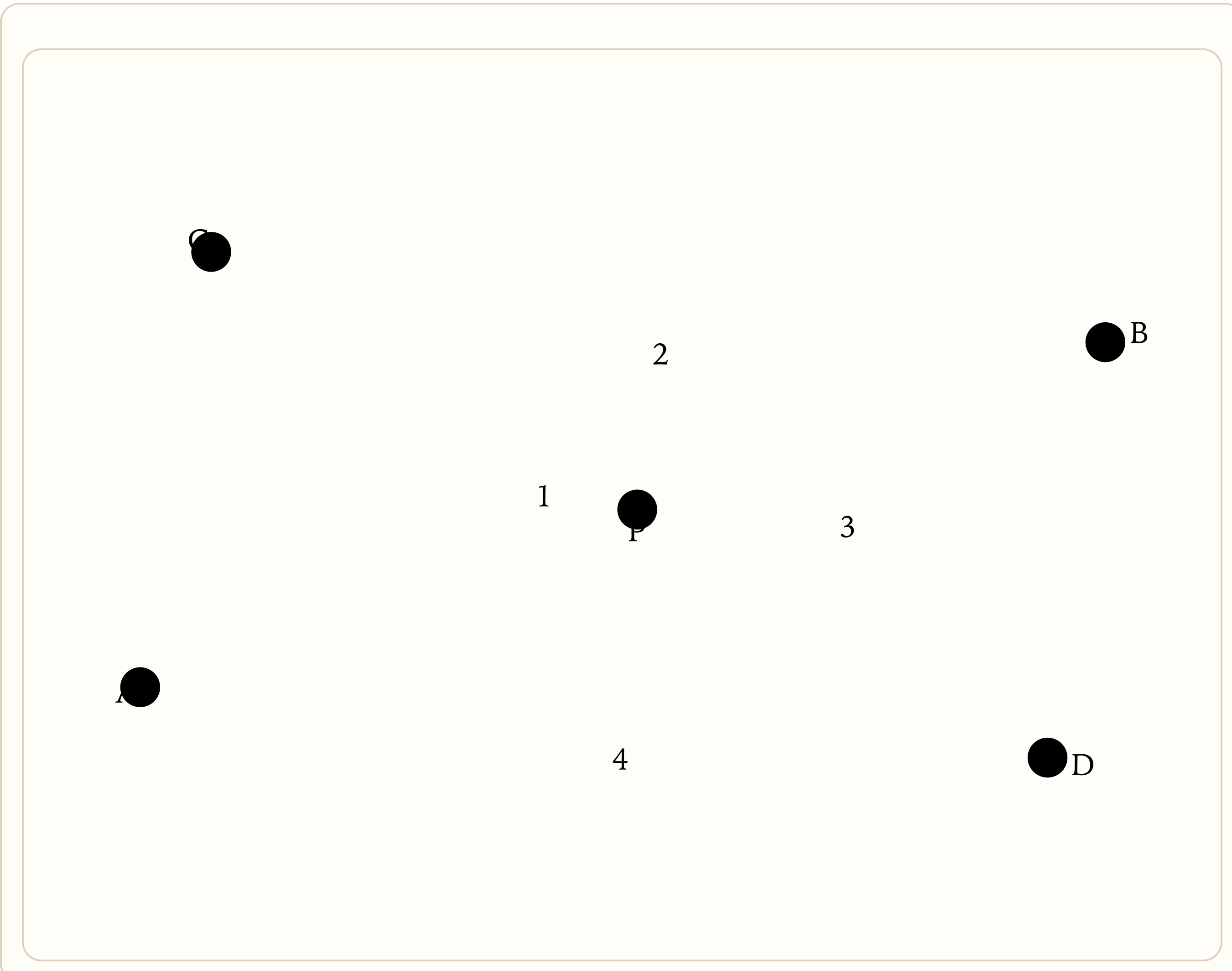
Vertical angles are equal

STATEMENT

When two straight lines cut each other, they make the vertical (opposite) angles equal.

GIVEN

Let AB and CD cut at P. Claim: vertical angles are equal — ∠1 = ∠3 and ∠2 = ∠4.



THEOREM 14

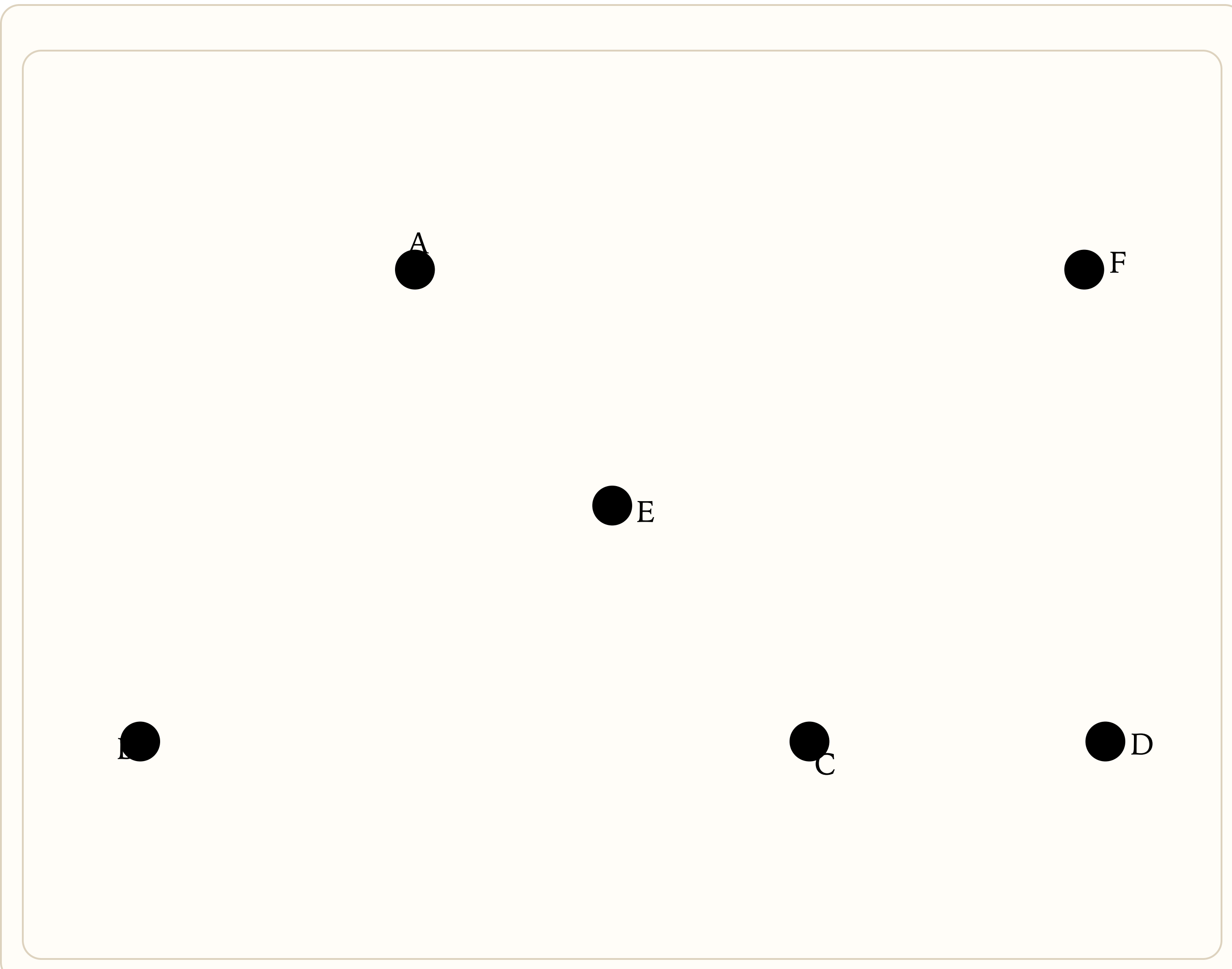
Exterior angle greater than remote interior

STATEMENT

If any side of a triangle is extended, the exterior angle is greater than either of the interior and opposite (remote) angles.

GIVEN

In △ABC extend BC to D. Claim: exterior ∠ACD > ∠BAC and ∠ACD > ∠ABC.



THEOREM 15

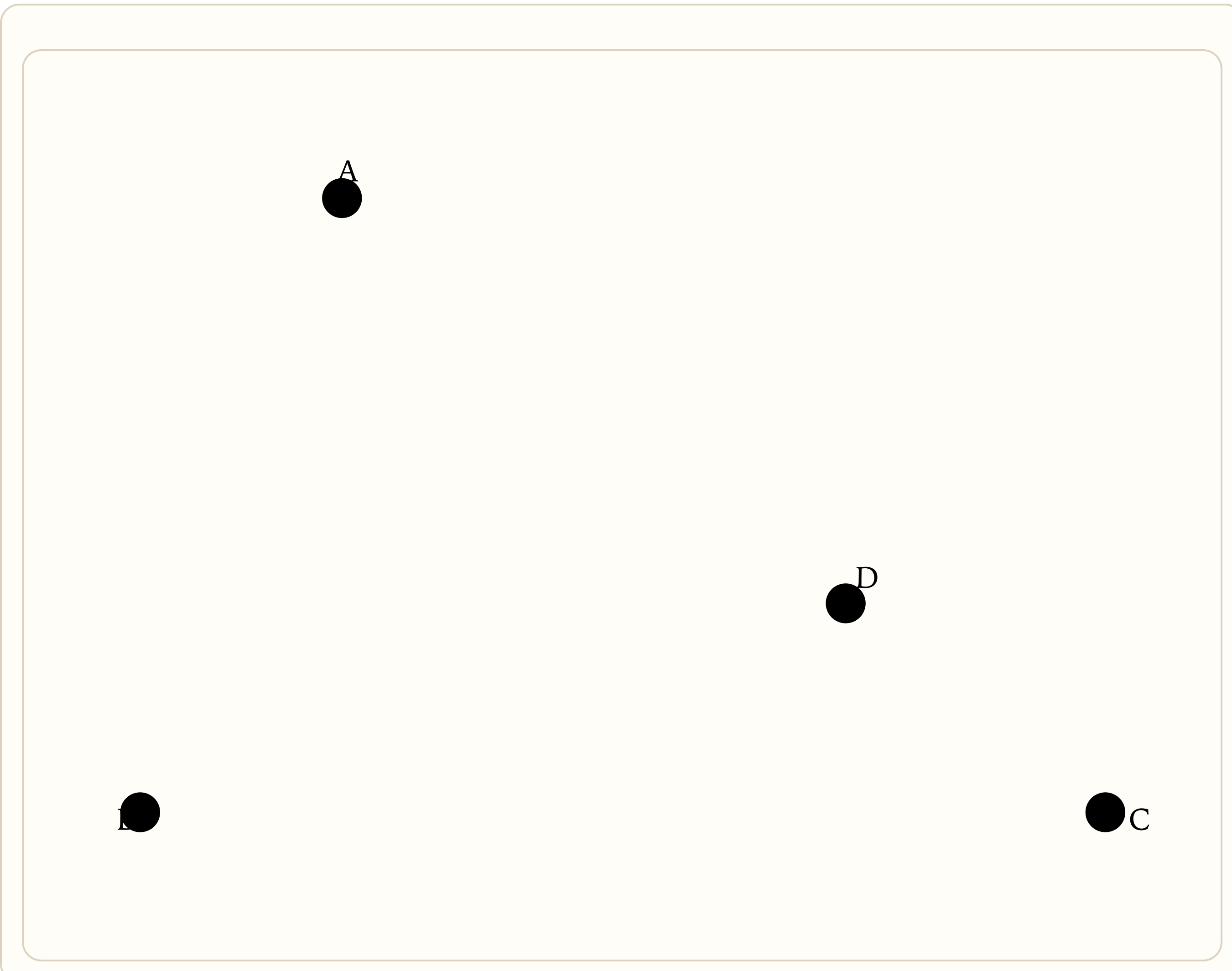
Greater side opposite greater angle

STATEMENT

In any triangle, a greater side has opposite to it a greater angle.

GIVEN

In $\triangle ABC$ suppose $AC > AB$. Claim: $\angle ABC > \angle BCA$.



THEOREM 16

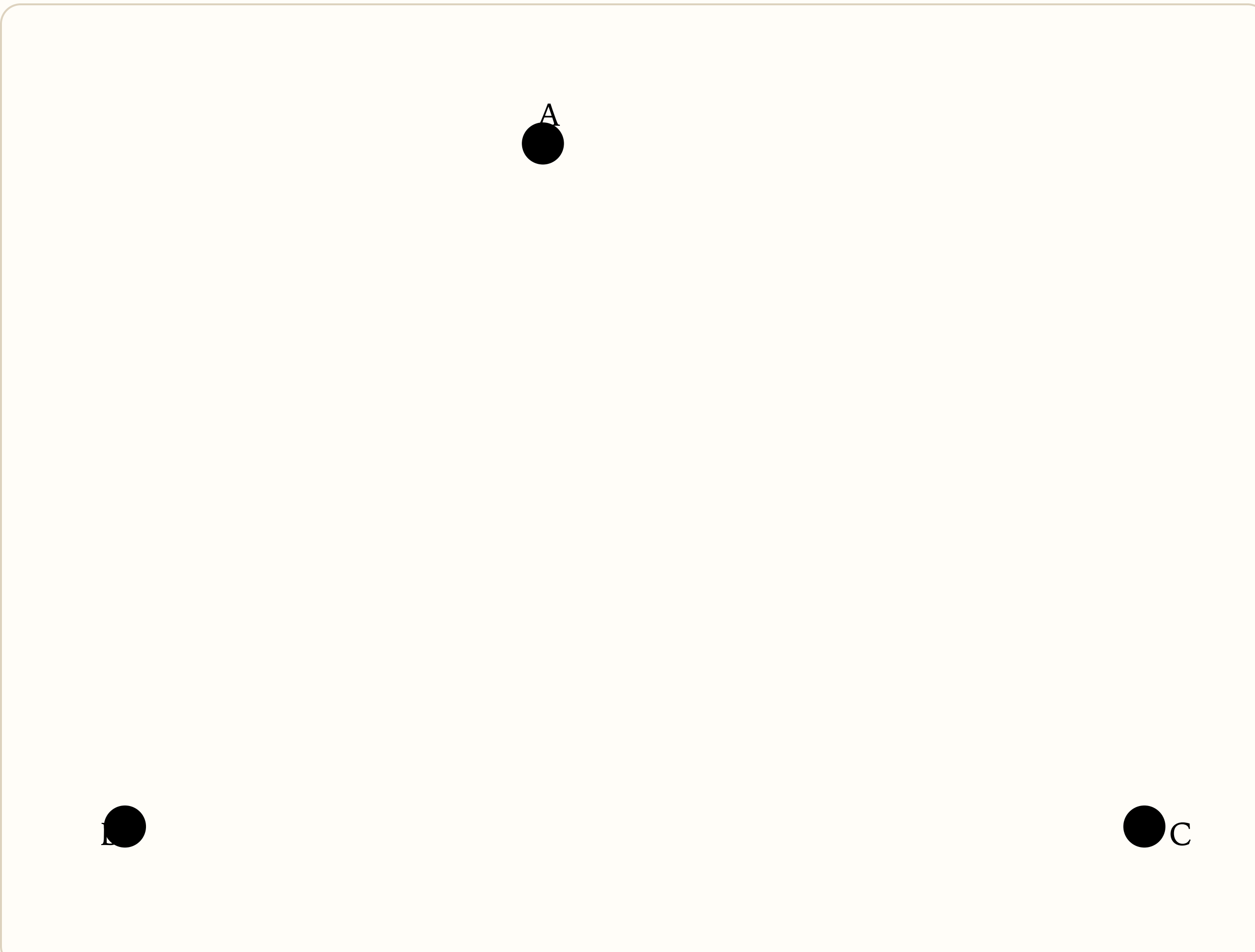
Greater angle opposite greater side

STATEMENT

In any triangle, a greater angle has opposite to it a greater side.

GIVEN

In $\triangle ABC$ suppose $\angle ABC > \angle ACB$. Claim: $AC > AB$.



THEOREM 17

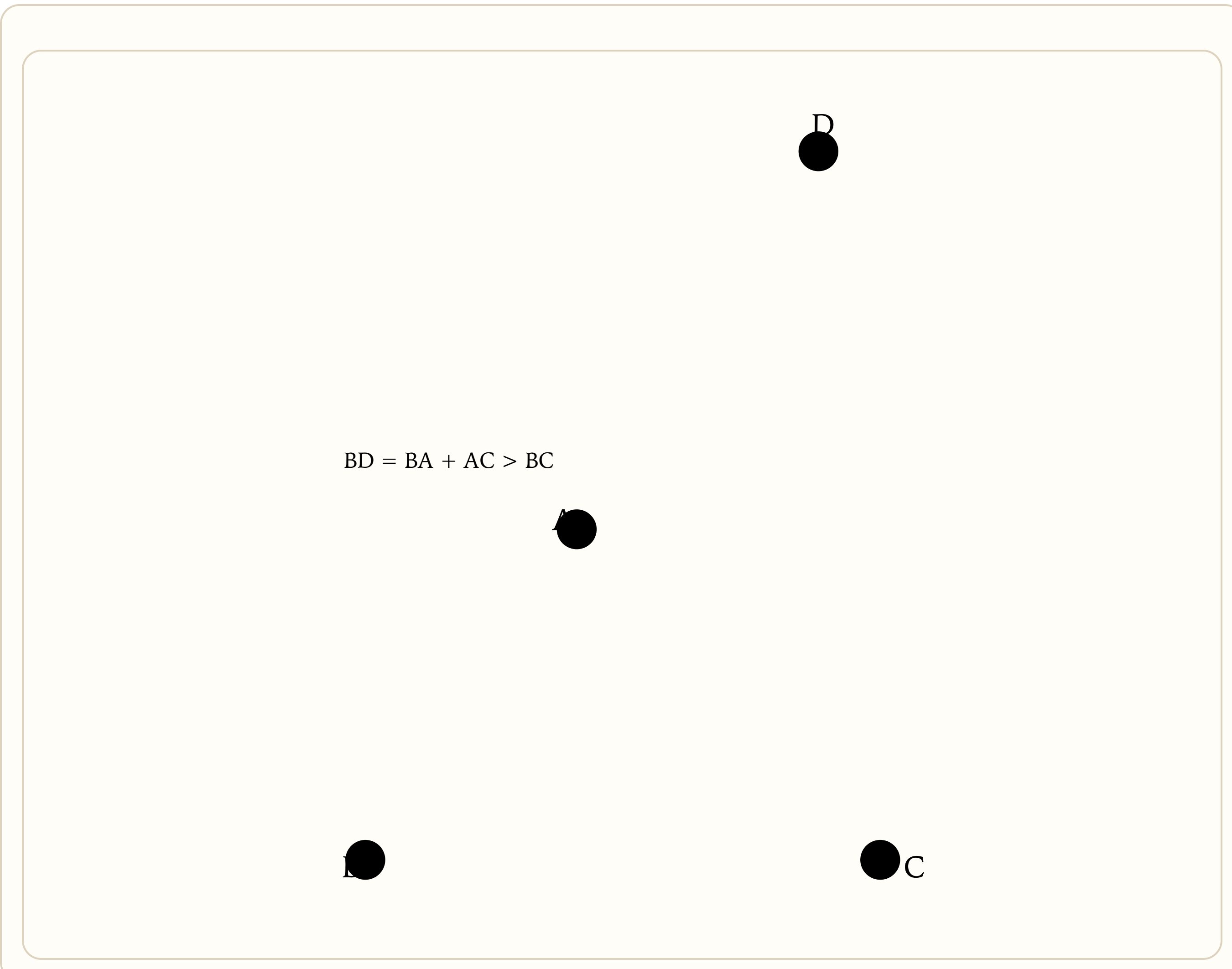
Triangle inequality

STATEMENT

In any triangle, any side is less than the sum of the other two sides, but greater than their difference.

GIVEN

In $\triangle ABC$ claim $BA + AC > BC$ (and cyclic).



THEOREM 18

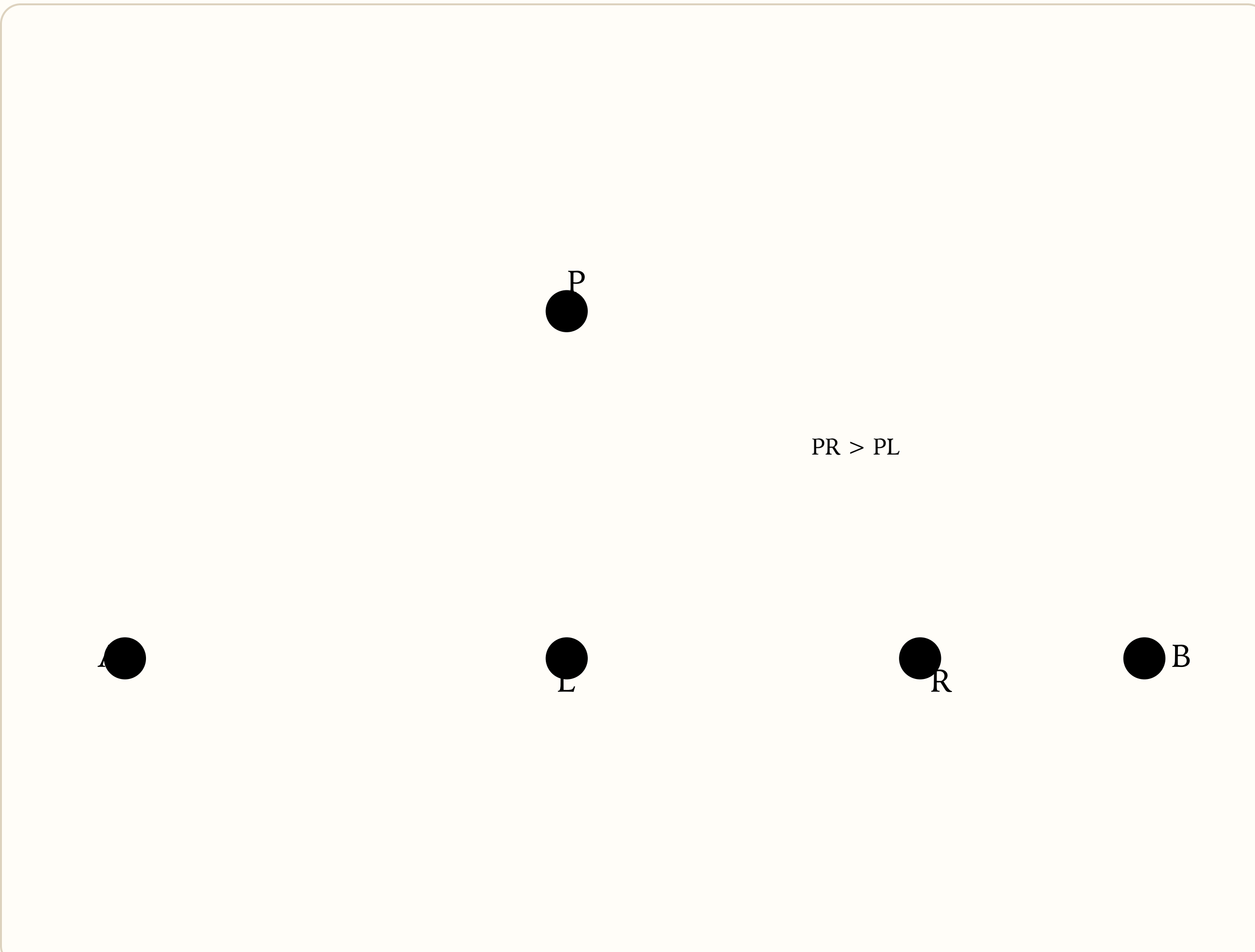
Shortest path to a line is the perpendicular

STATEMENT

The shortest distance from a point to a straight line is the perpendicular from the point to the line.

GIVEN

From point P drop perpendicular PL to line AB (Thm. 10). Let R be any other point on AB. Claim: PR > PL.



CONSTRUCTION 19

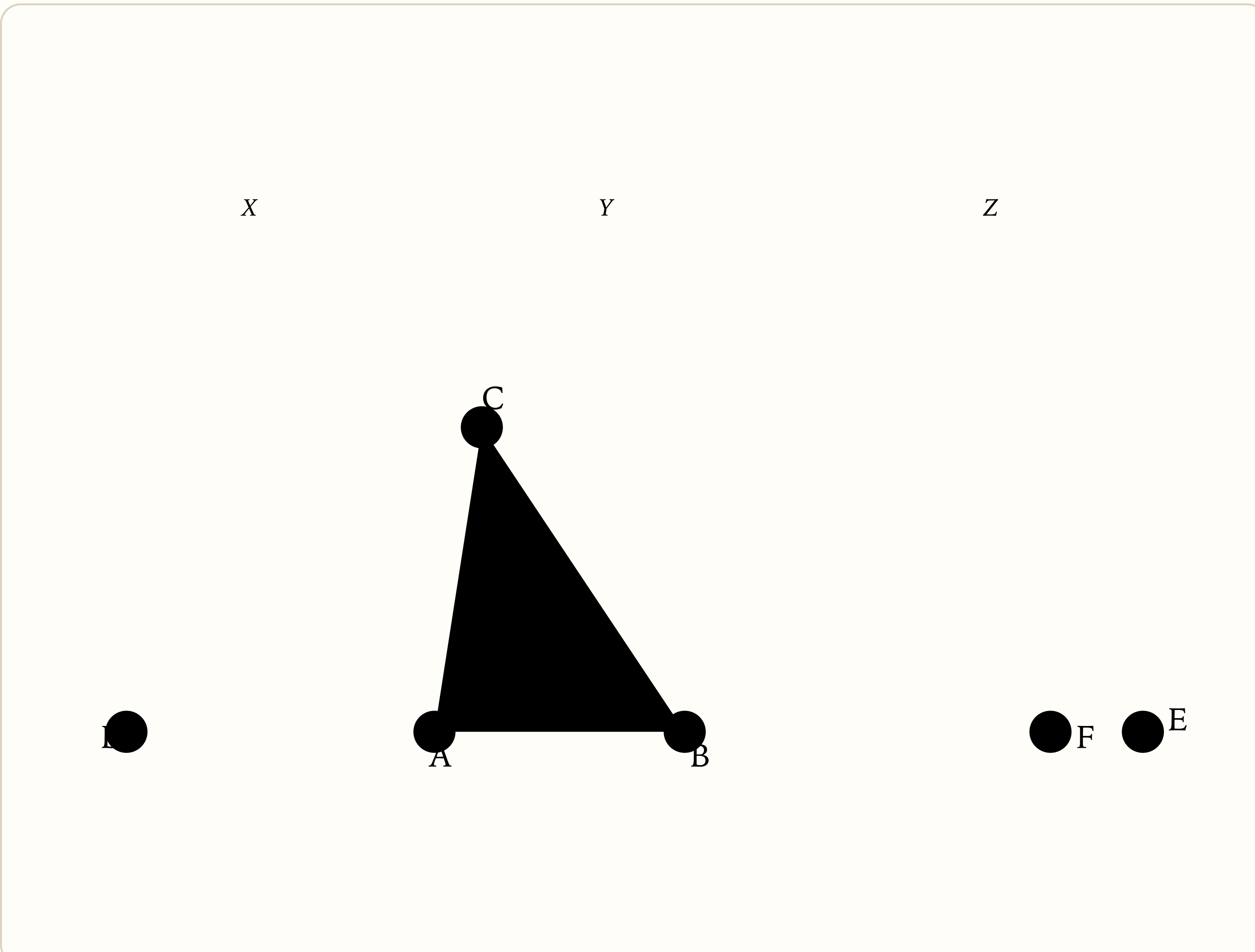
Construct a triangle from three sides

STATEMENT

Given three straight lines X, Y, Z such that any two together are greater than the third, construct a triangle with those side lengths.

GIVEN

Place X as DA and extend to E as needed; cut off AB = Y and BF = Z.



CONSTRUCTION 20

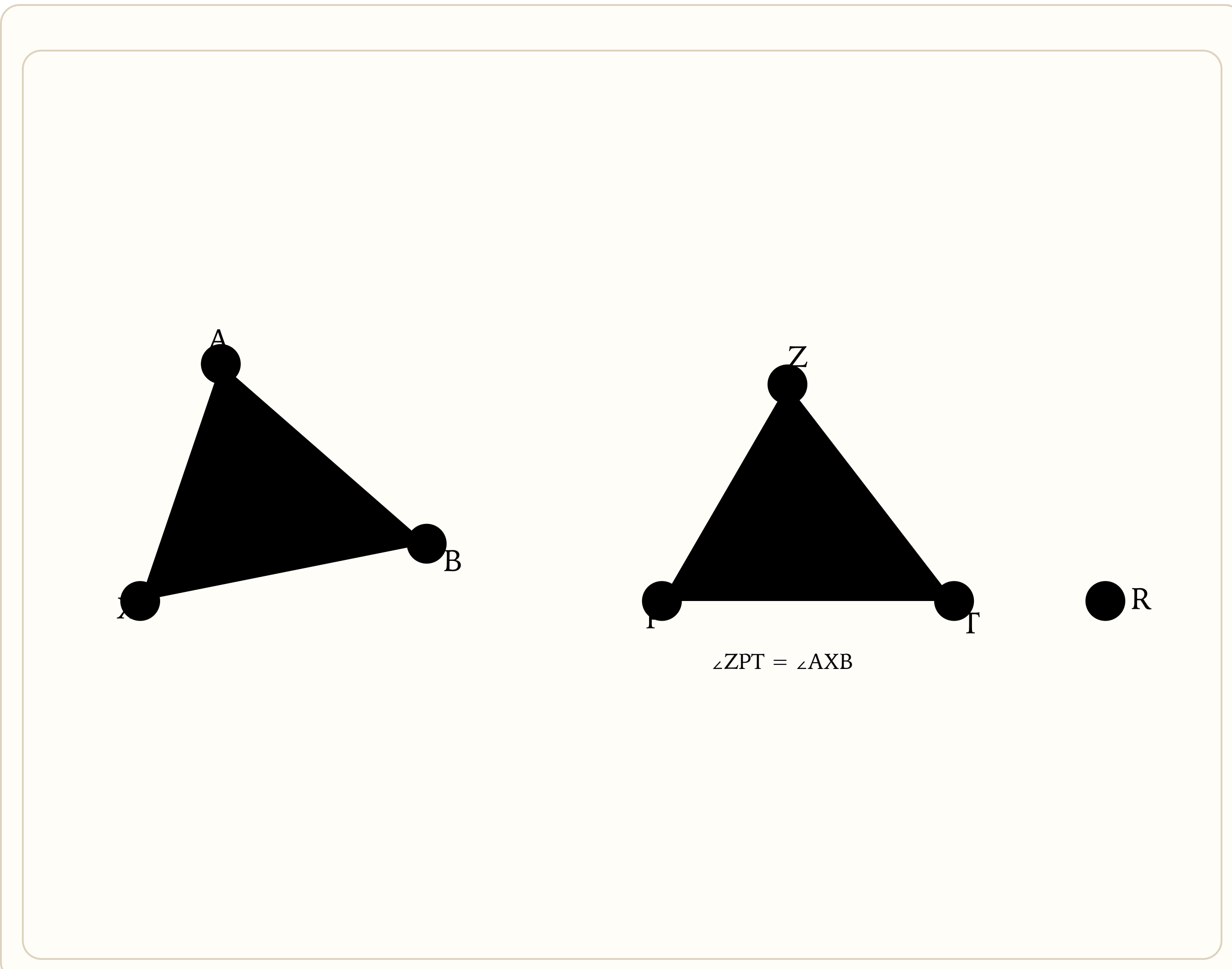
Copy an angle

STATEMENT

Given an angle X and a ray PR, construct an angle at P, with one side along PR, equal to angle X.

GIVEN

Pick points A and B on the legs of ∠X and join AB.



Angle-Side-Angle (ASA)

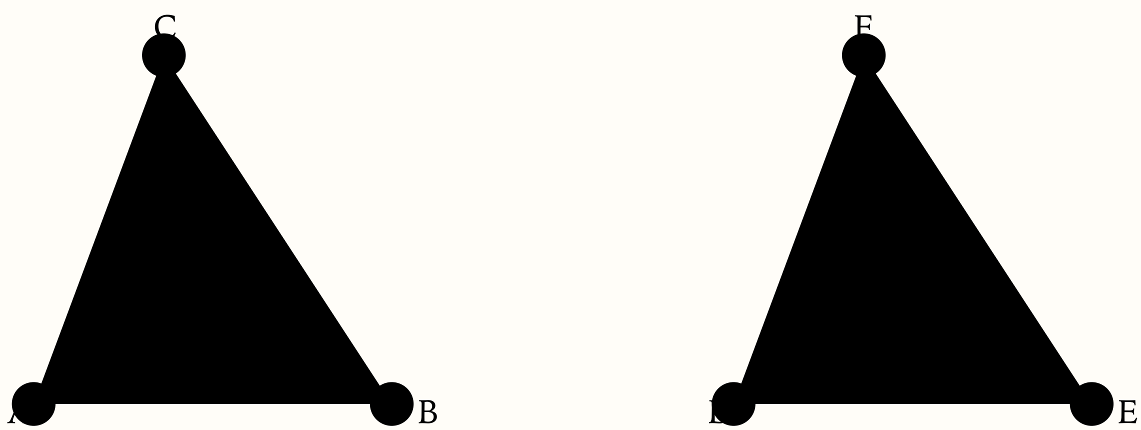
ASA Congruence

STATEMENT

If in a triangle two angles and the side joining them are respectively equal to two angles and the side joining them in another triangle, then all corresponding sides and angles are equal, and the triangles have equal area.

GIVEN

Given $\triangle ABC$, $\triangle DEF$ with $\angle BAC = \angle EDF$, $AB = DE$, $\angle ABC = \angle DEF$.



THEOREM 22

Angle-Angle-Side (AAS)

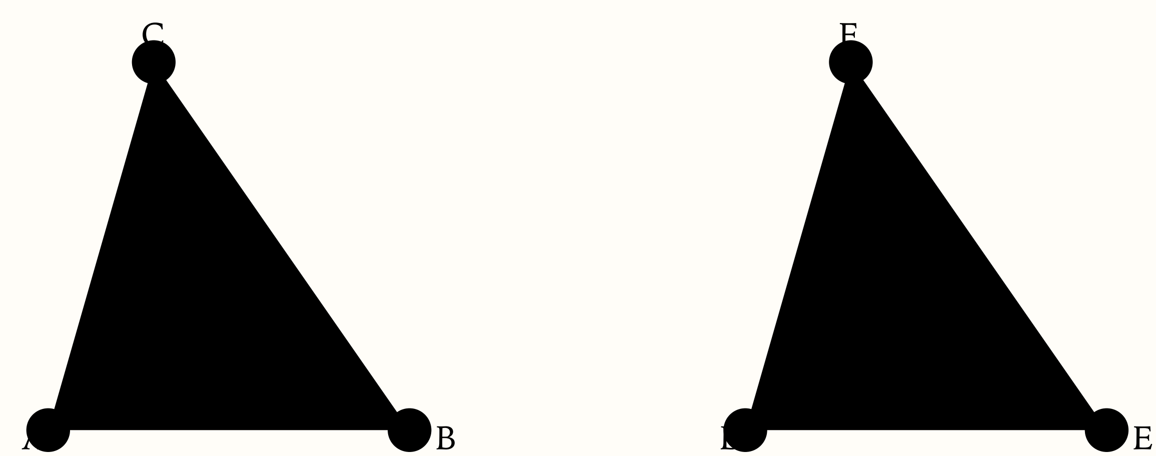
AAS Congruence

STATEMENT

If two triangles have two angles equal to two angles, and a side equal to a corresponding side opposite an equal angle, then they have all corresponding sides and angles equal, and equal areas.

GIVEN

Given ∠BCA = ∠EFD, ∠BAC = ∠EDF, and AB = DE (sides opposite a pair of equal angles, in Augros's setup).



THEOREM 23

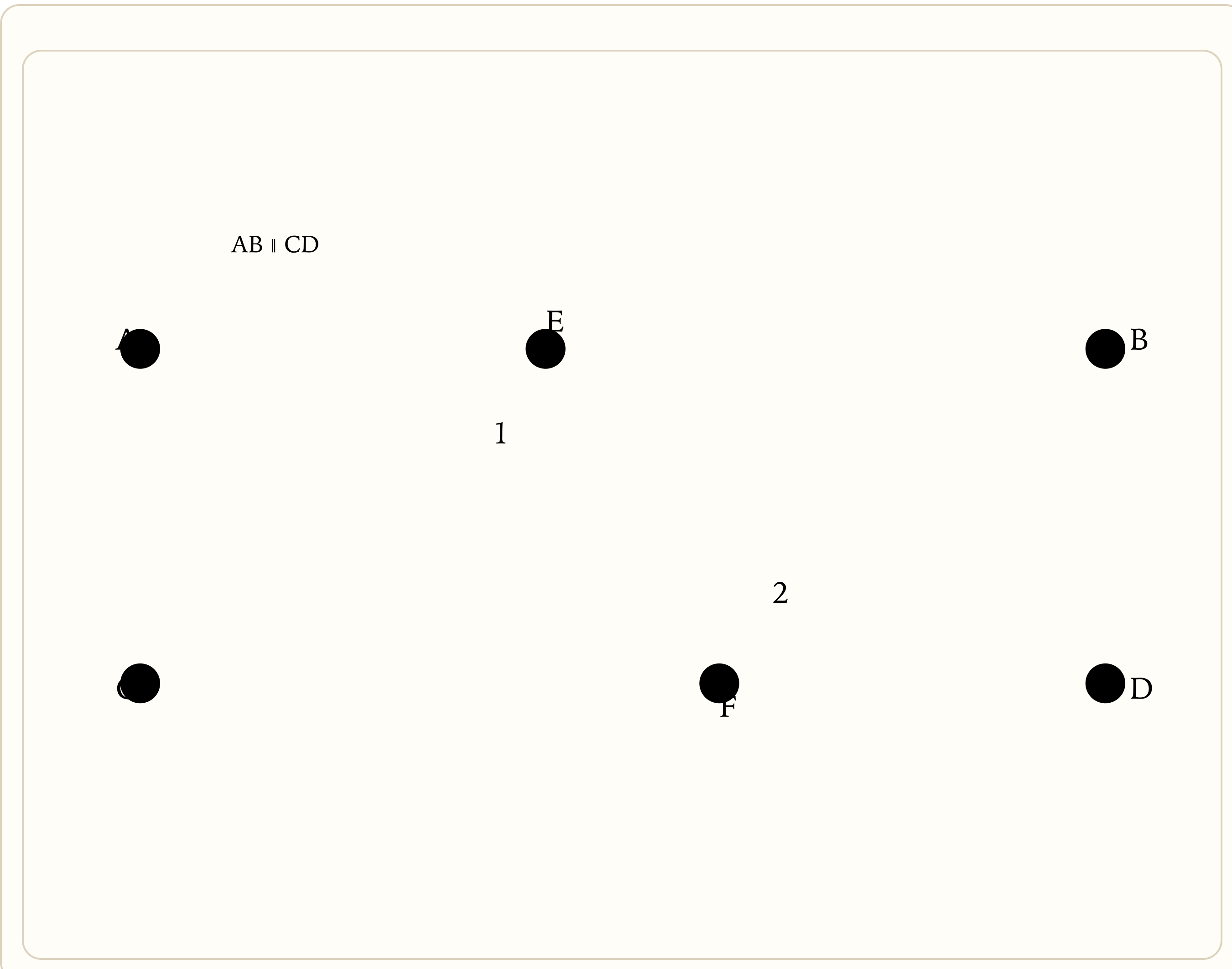
Equal alternate angles ⇒ parallel

STATEMENT

If two straight lines are cut by a third making the alternate angles equal, then the two straight lines are parallel.

GIVEN

Let EF cut AB and CD with alternate angles $\angle 1 = \angle 2$. Claim: $AB \parallel CD$.



THEOREM 24

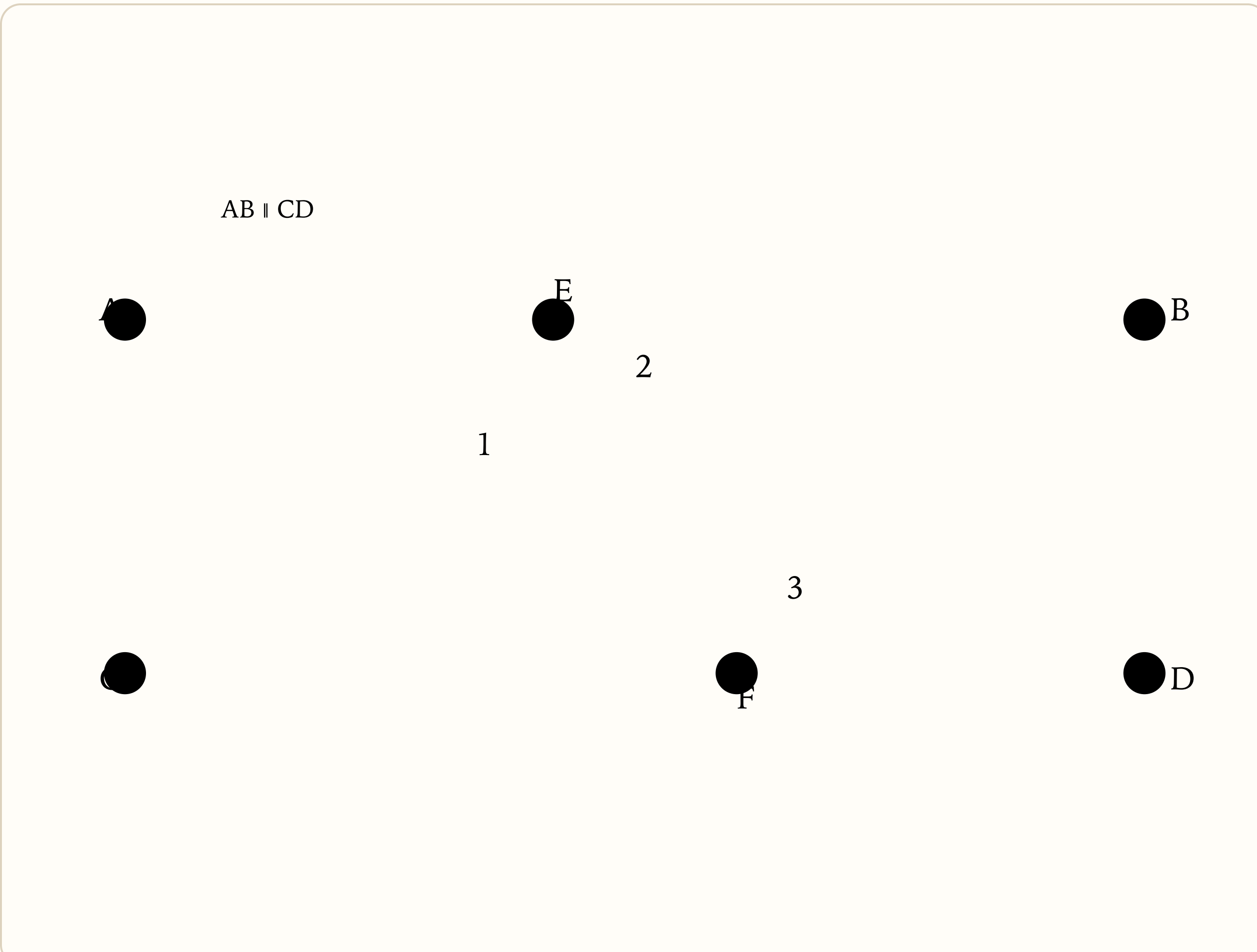
Co-interior angles supplementary ⇒ parallel

STATEMENT

If two straight lines are cut by a third making interior angles on the same side add up to two right angles, then the two lines are parallel.

GIVEN

Let EF cut AB and CD with ∠2 + ∠3 = two right angles. Claim: AB ∥ CD.



THEOREM 25

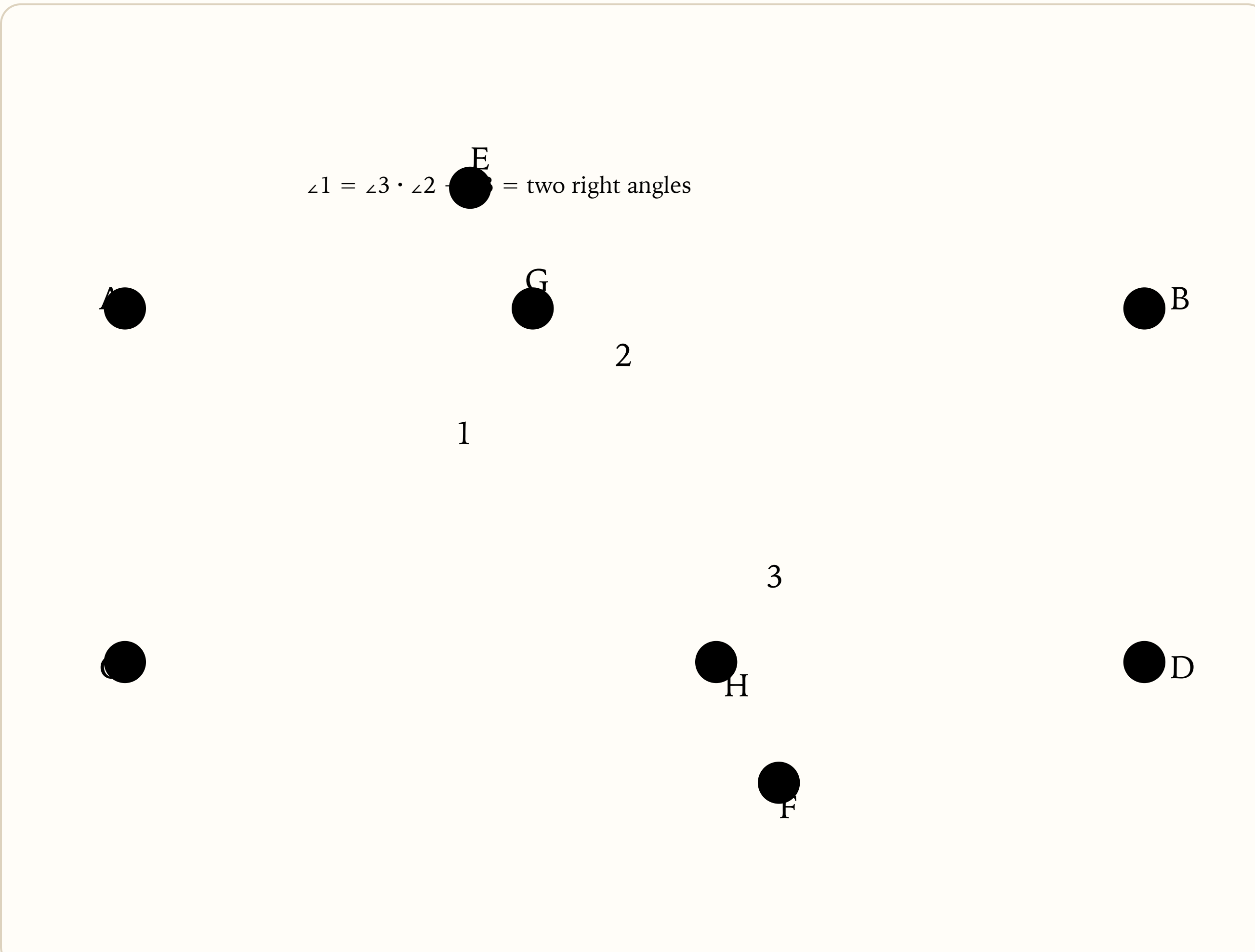
Parallel ⇒ alternate equal and co-interior supplementary

STATEMENT

If a straight line cuts two parallels, it makes alternate angles equal and interior angles on the same side supplementary (sum to two right angles).

GIVEN

Let $AB \parallel CD$ cut by transversal EGHF. Claim: alternate $\angle 1 = \angle 3$ and co-interior $\angle 2 + \angle 3 =$ two rights.



THEOREM 26

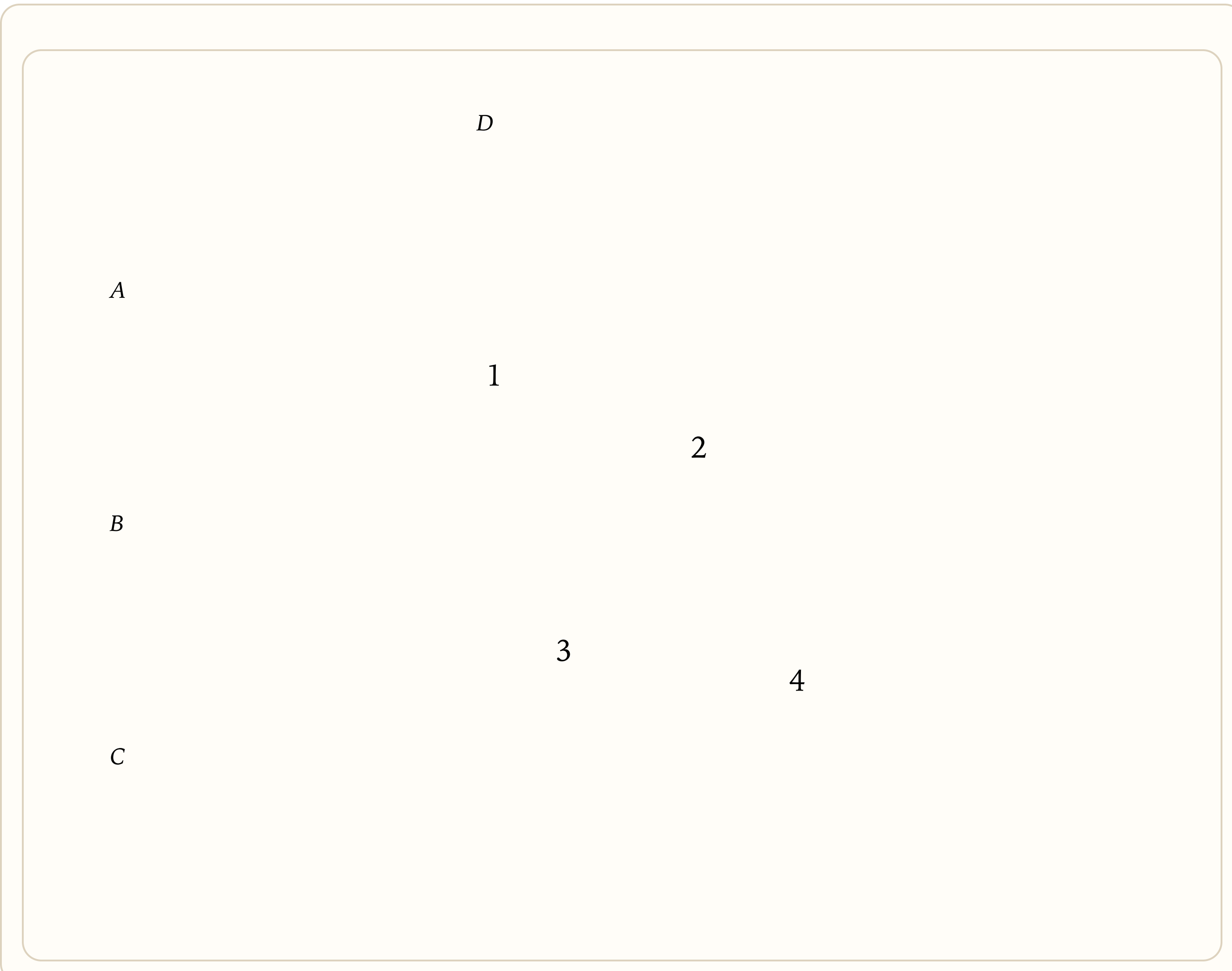
Transitivity of parallelism

STATEMENT

Straight lines parallel to the same straight line are parallel to each other.

GIVEN

Let $A \parallel B$ and $B \parallel C$. Claim: $A \parallel C$.



CONSTRUCTION 27

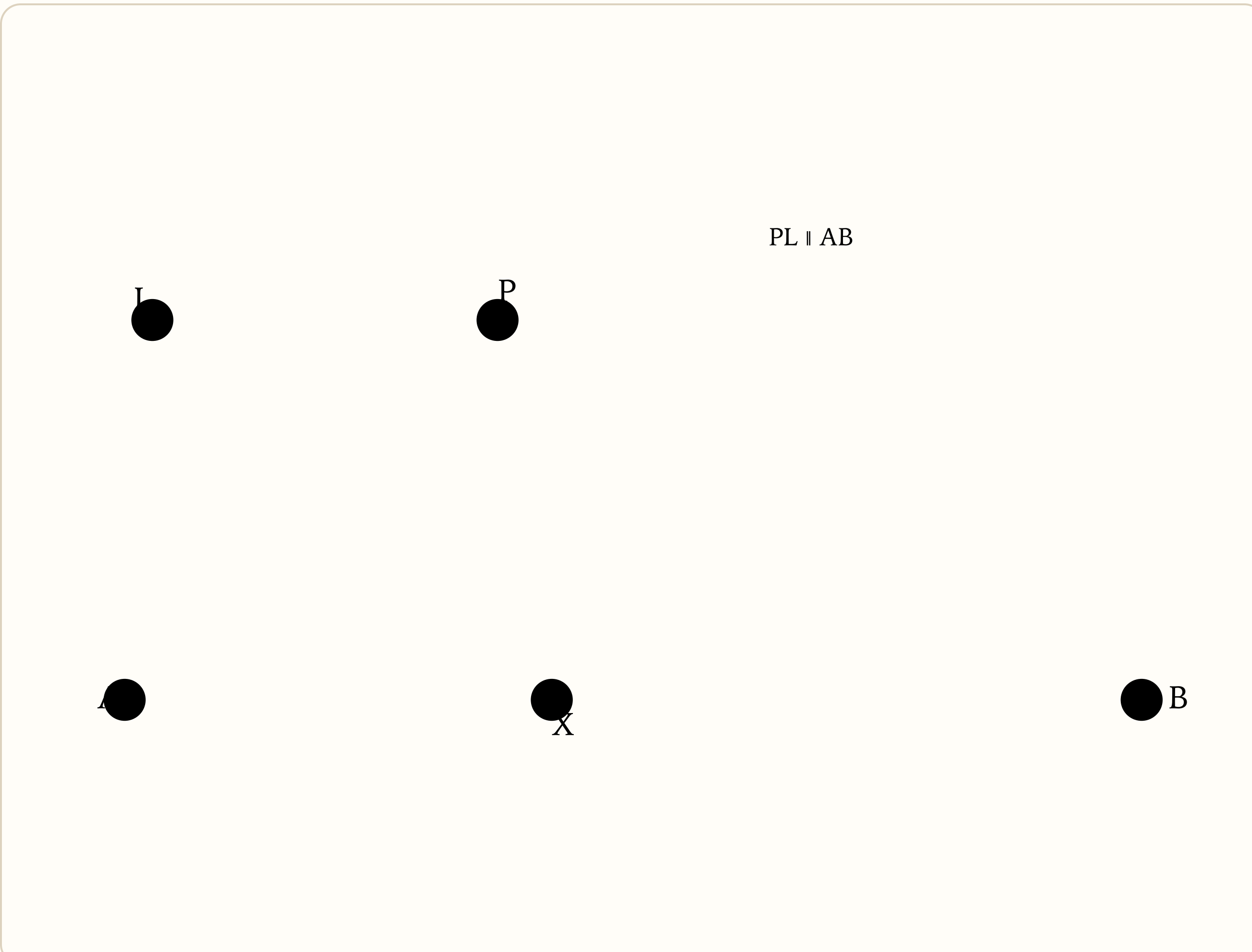
Draw a parallel through a point

STATEMENT

Given a line AB and a point P not on it, draw the line through P parallel to AB.

GIVEN

Pick any point X on AB; join PX.



THEOREM 28

Angle sum of a triangle

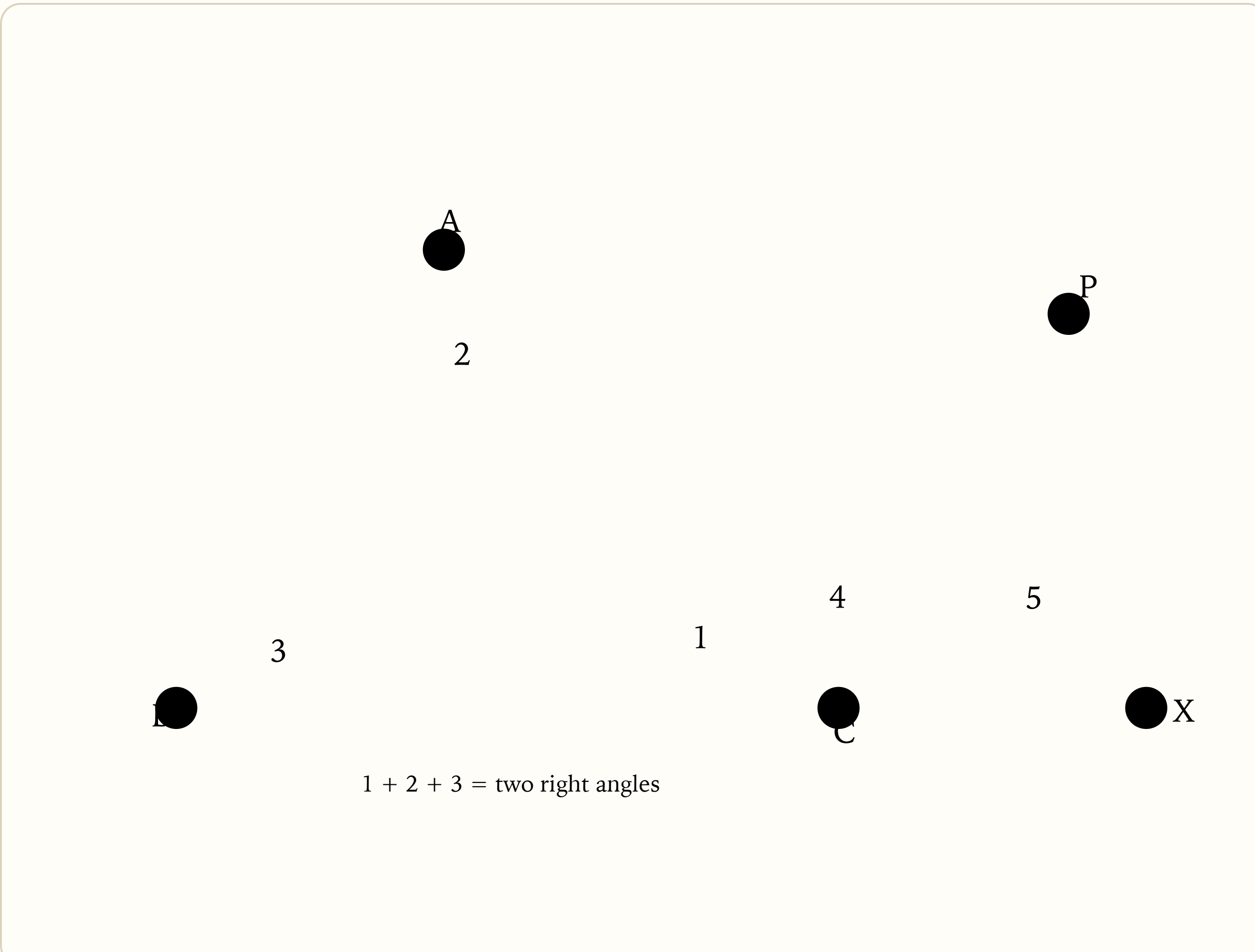
Triangular Angle-Sum Theorem

STATEMENT

In any triangle the three angles add up to two right angles, and an exterior angle equals the sum of the two remote interior angles.

GIVEN

In △ABC with angles 1, 2, 3, extend BC to X. Claim: exterior ∠ACX = 2 + 3, and 1 + 2 + 3 = two rights.



THEOREM 29

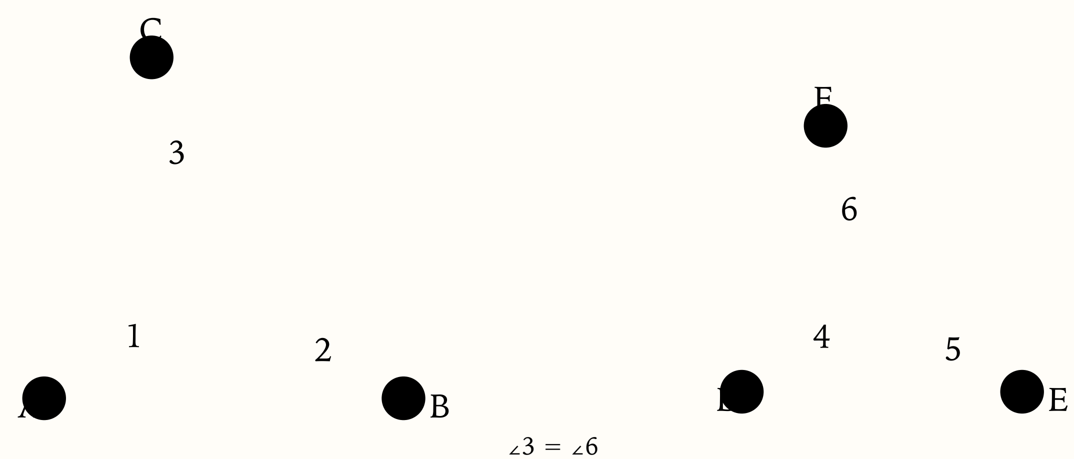
Third angle equal

STATEMENT

If two triangles have two angles equal to two angles, then the remaining angles are equal.

GIVEN

In △ABC and △DEF let ∠1 = ∠4 and ∠2 = ∠5. Claim: ∠3 = ∠6.



THEOREM 30

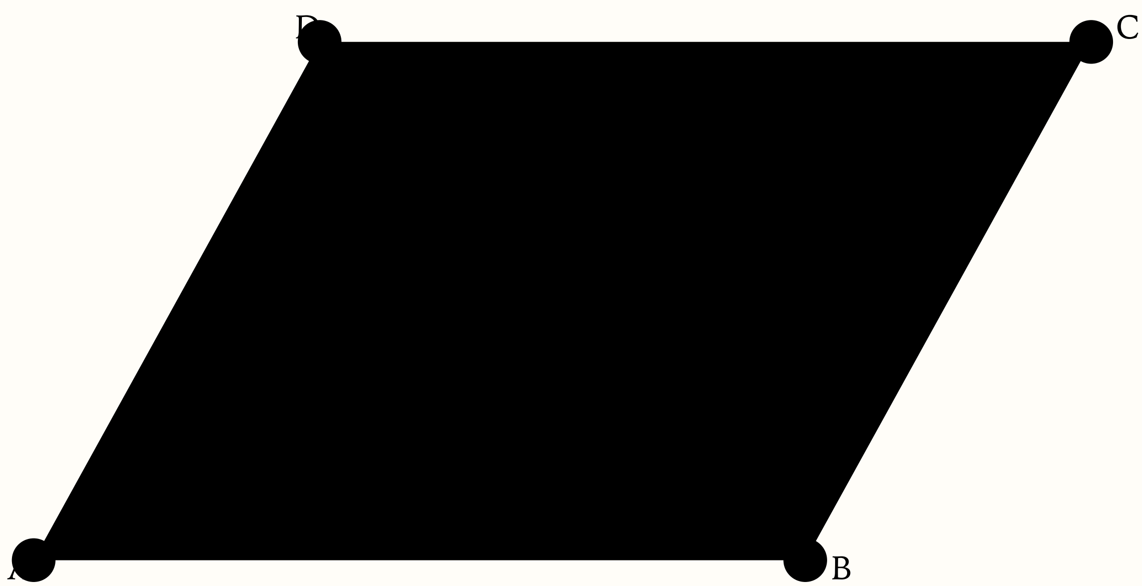
One pair parallel and equal ⇒ parallelogram

STATEMENT

If one pair of opposite sides in a quadrilateral are both parallel and equal, then the quadrilateral is a parallelogram (the other pair is also parallel and equal).

GIVEN

In quadrilateral ABCD let AB be parallel to DC and AB = DC. Join the diagonal BD.



AD ∥ BC and AD = BC

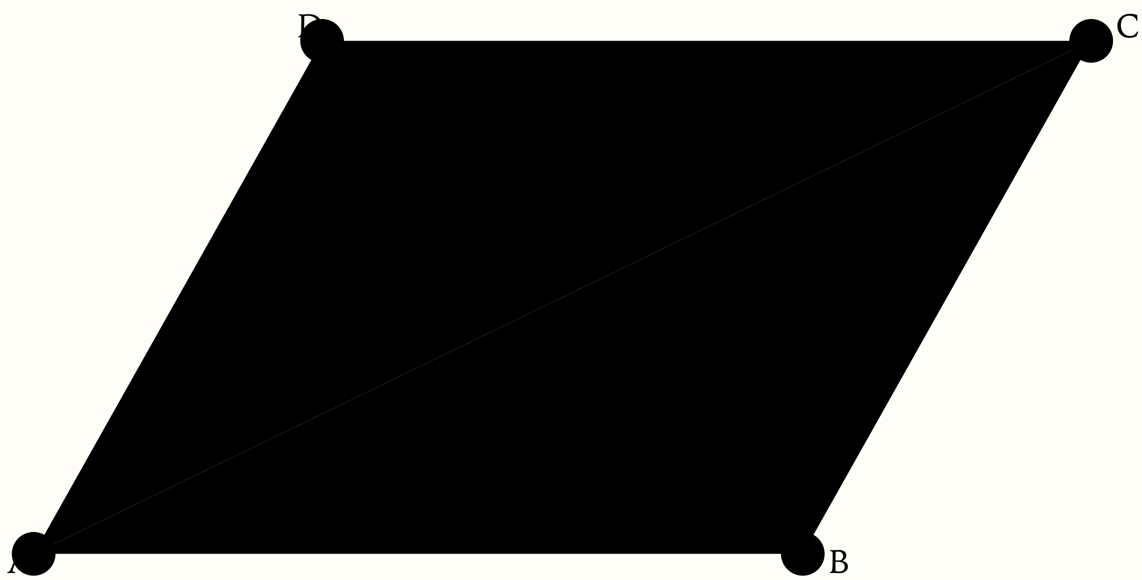
Properties of a parallelogram

STATEMENT

In any parallelogram, opposite sides and opposite angles are equal, and a diagonal bisects the area of the parallelogram.

GIVEN

In parallelogram ABCD, AB $\parallel$ CD and AD $\parallel$ BC. Draw diagonal AC.



THEOREM 32

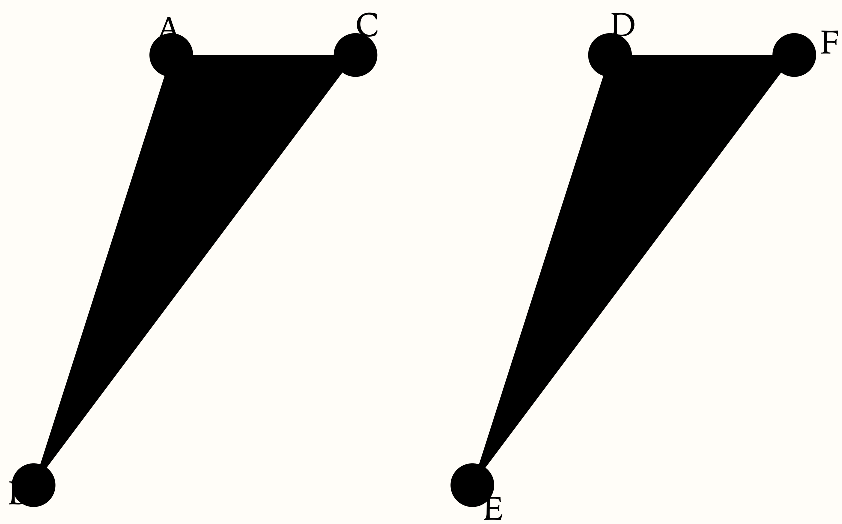
Parallelograms on the same base and in the same parallels

STATEMENT

Parallelograms on the same base and in the same parallels have equal areas.

GIVEN

Let parallelograms ABED and CBEF stand on base BE between parallels AF and BE.



parallelogram ABED = parallelogram CBEF in area

THEOREM 33

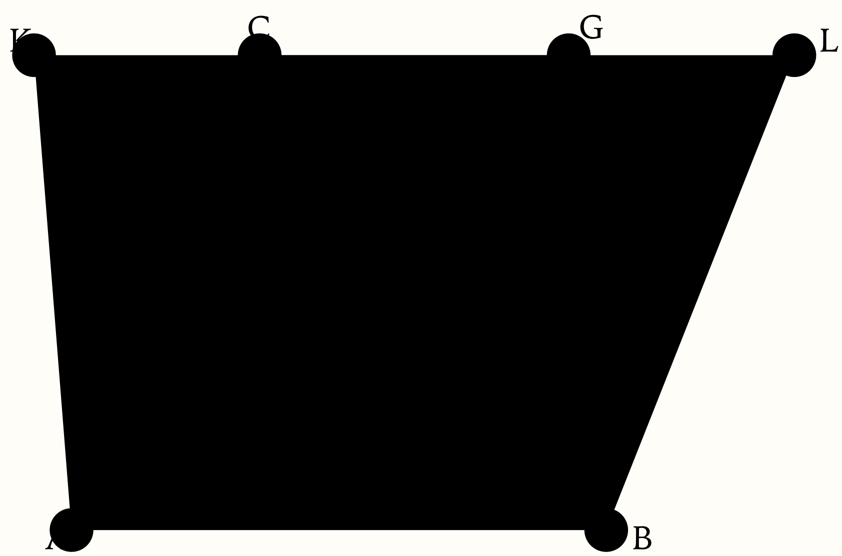
Triangles on the same base and in the same parallels

STATEMENT

Triangles on the same base and in the same parallels have equal areas; a triangle on the same base and in the same parallels as a parallelogram has half the parallelogram's area.

GIVEN

Triangles ABC and ABG stand on base AB in the same parallels.



$\triangle ABC = \triangle ABG$ in area

THEOREM 34

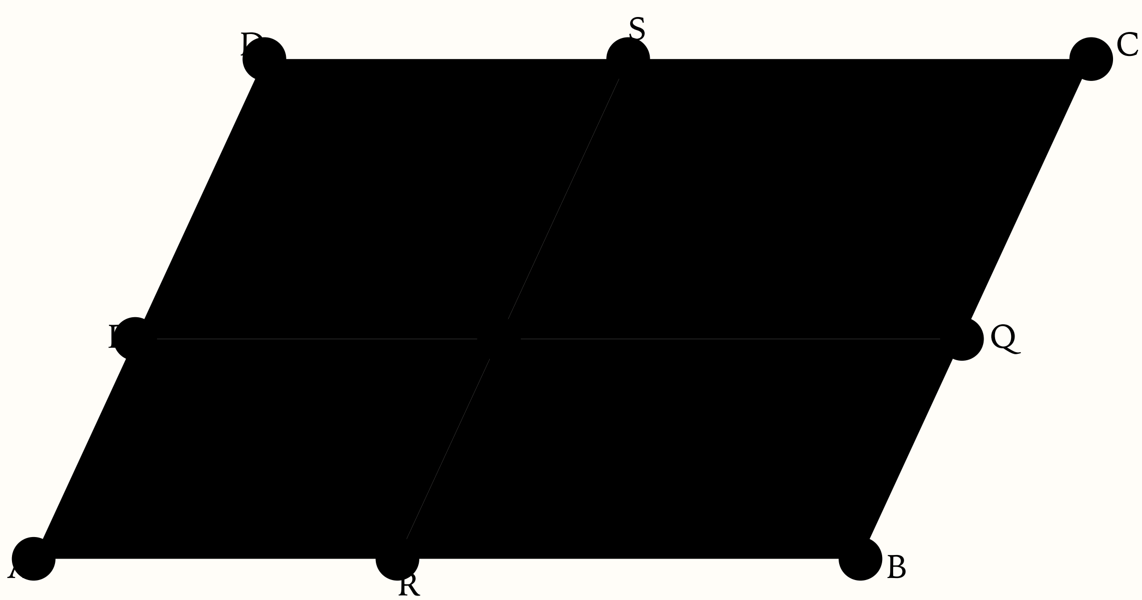
Complements of parallelograms about a diagonal

STATEMENT

In any parallelogram, the “complements” of the parallelograms about a diagonal are equal in area.

GIVEN

In parallelogram ABCD draw diagonal AC. Through a point K on AC draw lines parallel to the sides, forming small parallelograms about the diagonal and two complementary parallelograms (1) and (2).



complement (1) = complement (2)

CONSTRUCTION 35

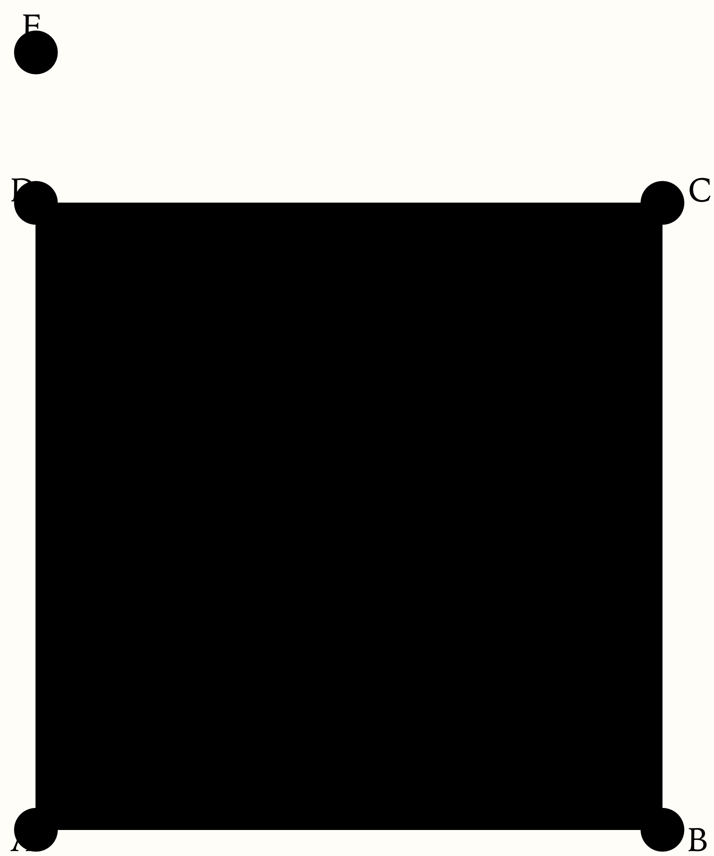
How to make a square

STATEMENT

Given a straight line AB, construct a square having AB as one side.

GIVEN

Erect AE perpendicular to AB and cut off AD = AB.



THEOREM 36

The Pythagorean Theorem

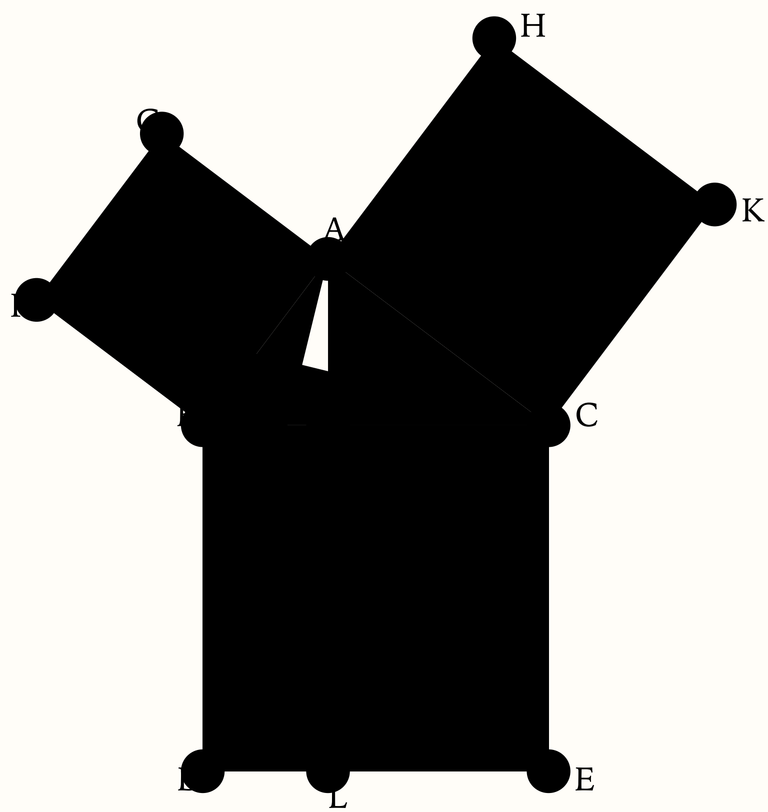
Pythagorean Theorem

STATEMENT

In any right triangle, the square on the hypotenuse is equal to the sum of the two squares on the remaining sides.

GIVEN

Right triangle ABC with right angle at A; squares ABFG on AB, ACKH on AC, BCED on hypotenuse BC.



square BCED = square ABFG + square ACKH

THEOREM 37

Converse of Pythagoras

STATEMENT

If the square on one side of a triangle equals the sum of the squares on the other two sides, then the angle opposite that side is a right angle.

GIVEN

In $\triangle ABC$ let the square on AB together with the square on AC be equal to the square on BC. Claim: $\angle BAC$ is a right angle.